Reviews or Ratings: Quantifying Information Loss from Coarsening*

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This paper studies the optimal design of a review system that elicits information from reviewers who willingly submit reviews. A platform, seeking to learn an unknown state, trades off between the informativeness and frequency of reviews. Finer reviews provide more information per review, but their complexity leads reviewers to submit them less frequently. Reviewers' signals capture their realized utility, which depends on the state of the world and an idiosyncratic individual component. I characterize the optimal review system. I apply this characterization to study how taste heterogeneity of reviews impacts the performance of review systems. When consumers are more homogeneous, coarse ratings perform worse. Moreover, when homogeneity is large, the optimal binary rating is asymmetric. Regardless of the degree of homogeneity, when the optimal threshold is used a binary rating is preferred to a full signal if it is submitted at least 3.25 times as frequently.

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1 Introduction

Transmitting information is costly. Yet, economic agents rely on others' information. In many settings, it is infeasible or inappropriate to compensate individuals for sharing their information. In some cases this is due to concerns that compensation may bias the information that agents provide (e.g., Amazon prohibiting vendors from paying consumers for reviews). In others, compensation is simply not a viable option (e.g., a school eliciting students' course reviews). In these settings, the collector of information is reliant on the goodwill of agents providing their own information. Due to this tension—a principal wants information, but lacks a way to directly incentivize agents to provide it—collectors of information should reduce barriers to sharing information.

One natural way to reduce frictions in information-sharing is to ask for less information. Intuitively, reducing the number or complexity of questions that a reviewer must answer increases the likelihood that they will complete the review. However, this method of reducing frictions is not without a major drawback: the simpler review will elicit and thus provide less information.

This paper's main contribution is explicitly modelling and quantifying this trade-off. I show when a platform prefers to elicit coarsened information, and what coarsened information is optimal. I propose a model where reviewers have a signal, and a principal (platform) acquires their information by asking them to submit a review. Eliciting more information from reviewers makes them less likely to submit their signal. The platform must trade off between more frequent and better information. I quantify this trade-off when the number of potential reviewers is large. I show that the principal's preference over different review systems is independent of its decision problem, but is dependent on the distribution of reviewers' signals. Independence across decision problems implies that the platform can apply the same methods across products and settings without having to closely monitor other aspects of the environment. This is especially relevant for large organizations, where the designers of review systems are not necessarily those using the information. Dependence on reviewers' information means that there is a strong incentive for the platform to understand how the distribution of signals affects the trade-off between frequency and quality. I provide conditions under which coarsened reviews lead to minimal loss of information, and then use these conditions to justify why certain types of review systems are prevalent. In particular, my results predict the optimality of positively-skewed review systems and the prevalence of binary ratings systems.

In the model, a platform elicits information from a pool of agents (reviewers) who each have a signal that is informative of the underlying state of the world (for example, previous consumers of a good). The platform chooses a mapping from signals into reviews. Reviewers observe the mapping, and each chooses whether or not to submit their signal. This choice depends on a private willingness to review and the number of possible reviews they can leave: reviewers are less likely to leave a review when they must decide between many possible options.

Motivated by online reviews, I focus on the case when there is a large number of reviewers. In the limit, the state is learned, and so what matters is the *rate* at which the state is learned by the platform. This rate depends on both the average informativeness of a single review and the probability that a reviewer leaves a review. Since this rate is independent of the platform's decision problem, this setting admits a decision problem-free ranking of review systems.

Technically, I leverage large deviations techniques in order to characterize this rate, for each review system. These tools allow me to characterize the trade-off between frequency and quality of a review. The ranking is separable across this trade-off, enabling independent analysis. Since the rate of review can easily be computed empirically, I focus on characterizing the informativeness of a single review for different review systems. In particular, I characterize the information lost in a coarse review in terms of how much more frequently it must be submitted than a full review in order to be preferred by the platform.

A key characteristic of online review platforms is that a single review is almost uninformative: individual taste differences mean that, on average, any one reviewer's experience is a very noisy signal about the underlying quality of the product. In this "scarce information" context, I characterize how much information is lost in terms of how much more frequently the coarse review to be submitted relative to a full-signal review for the platform to prefer the coarse review. This in turn allows me to determine what is the optimal review system.

In practice, many discrete rating systems degenerate to a review system resembling a binary system. It has been well-documented (see Hu, Zhang, and Pavlou (2009)) that many 5-star review systems follow a bimodal distribution. Even in this simple setting, the platform must determine how much information is lost when a simple positive/negative review is used as well as determine the optimal definition of "negative". I apply the results on quantifying information loss to answer these questions. Specifically, I study how the answers are influenced by the dispersion of taste shocks in the reviewers' signals. Here, larger dispersion reflects more heterogeneous experiences within the population for the product being reviewed.

A key insight is that a more homogeneous population of reviewers implies that even when information is scarce, some reviewers have very informative signals. When coarse reviews are used, these very informative signals are mixed with a much larger mass of uninformative signals, drowning out their informational content. On the other hand, when reviewers are very heterogeneous, the variance of signal informativeness is small. As a result, less information is lost in a simple review when the population is heterogeneous.

Another key result that the definition of a negative review is extremely important for minimizing information-loss. When reviewers are sufficiently heterogeneous, the optimal binary review system is simple: the platform symmetrically asks reviewers if their experience was good or bad. However, when homogeneity is large, the platform uses an asymmetric review: it isolates either "horrible" or "amazing" experiences. For very homogeneous populations, information loss of the "good"/"bad" binary review relative to the full review is unbounded. However, the optimal review bounds information loss. For any degree of homogeneity, if a reviewer is 3.25 times as likely to submit a binary review as their full signal, the optimal binary system outperforms the full review.

These results suggest that even extremely coarse reviews can be constructed such that they retain a relatively large amount of information. The framework also helps explain why the ratings systems that are observed in practice are implemented, even when they seem counter-intuitive. The stark asymmetry in review systems used by platforms may indeed be optimal, despite the appearance of sub-optimality. For instance, reviewers on many platforms have converged to positively skewing reviews (e.g., many users default to 5-stars on Uber and Airbnb). I show that if reviewers exhibit less negative than positive heterogeneity (i.e., they agree more on what constitutes a negative experience than a positive one), it is always optimal for the platform to isolate "horrible" experiences, while grouping all other experiences together. This results in a much larger number of positive reviews than negative reviews, even when the state is low.

The outline of the paper is as follows. Section 2 presents and discusses the model. Section 3 formalizes the trade-off between review informativeness and reviewers' frequency of review and explicitly characterizes optimal review systems when information is scarce. Section 4 applies these results to study the impact of heterogeneity on review systems. Section 5 discusses several extensions and Section 6 discusses and concludes. All proofs are included in the Appendix.

1.1 Related Literature

This paper is closely related to the literature studying learning dynamics in recommendation and review systems.

A large strand of this literature focuses on social learning dynamics. Ifrach et al. (2019) studies a standard social learning setting with binary reviews. Besbes and Scarsini (2018) looks at a similar setting, and focuses on when boundedly-rational agents use of summary statistics of reviews leads to eventual learning. Most similar to my work is Acemoglu et al. (2022), which studies learning rates for different review systems in the context of social learning. Although both that study and this work are interested in learning rates, there are several major differences. In particular, instead of focusing on the effect of social learning, I focus on quantifying the trade-off between different review systems. Additionally, I focus on a setting where a key difference in different review systems is the rate at which reviews are left. In the setting of Acemoglu et al. (2022), where reviews are automatically submitted, a finer review system is always preferred to a coarser system; in my setting this is not the case.

There is also a related literature that takes a different approach to studying recommendation and review systems. For instance, Che and Hörner (2018) studies a platform which aims to understand the quality of a product by "pushing" it to consumers. The signal structure in that model is perfect good news, but the platform cannot prove the good news to consumers. A different approach is taken in Garg and

Johari (2019), which uses large deviations techniques to optimally differentiate many underlying states by controlling the rate at which positive reviews are given for each quality level. A major difference between this work and Garg and Johari (2019) is that there, the designer has full control over the information system, whereas in my case the designer is constrained by the information of reviewers.

My work is also informed by the empirical literature on ratings. Looking at the common five-star system, it has been well documented that, across platforms, the distribution of submitted ratings is positively-skewed and bimodal (see Chen, Yoon, and Wu (2004) and Hu, Zhang, and Pavlou (2009)). Hu, Pavlou, and Zhang (2017) studies how this distribution is due to a self-selection effect in the population of reviewers. Fradkin, Grewal, and Holtz (2021) and Fradkin and Holtz (2023) study possible interventions to improve the rate and quality of review submission. These works find that while these interventions (including directly incentivizing reviews) can increase the rate at which reviews are submitted, they do not tend to improve the overall informativeness of the review system, highlighting the need for careful design of review systems. For a survey of empirical work aiming to understand review systems, see Magnani (2020). Recent empirical work has also provided alternative reasons to use coarse review systems. For instance, Botelho et al. (2025) recently showed that moving from a five-star system to a binary rating decreased racial discrimination in an online platform that matches workers with customers. My work compliments this literature by providing an information-theoretic argument for using simple review systems, and suggesting that skewed reviews may be optimal.

My paper contributes also to the literature studying and applying rates of learning to understand economic contexts. This literature is broad, as learning rates have use in many settings. For instance, Rosenberg and Vieille (2019) and Hann-Caruthers, Martynov, and Tamuz (2018) study how quickly actions converge in a social learning setting, and Harel et al. (2021) and Dasaratha and He (2024) study the impact of networks on social learning.

Technically, this paper applies large deviations techniques to study rates of learning. There is a developed statistical literature that uses similar techniques to compare statistical experiments. Chernoff (1952) provides foundational results in the context of hypothesis testing. Torgersen (1981), which develops a generic ordering over different statistical experiments for finite decision problems when the number of signals is large, extends the ordering of Blackwell (1951). Large deviations techniques have been applied in several cases in the economics literature. Moscarini and Smith (2002) studies more sensitively the rate of learning, in order to determine characteristics of the demand for information. Frick, Iijima, and Ishii (2024b) ranks different misspecifications by the rate at which they slow learning.[1] Mu et al. (2021) develops a generalization of the ordering of Blackwell (1951) in the context of many signals. Finally, Fedorov, Mannino, and Zhang (2009) looks at similar trade-offs to those that I study in the context of hypothesis testing.

2 Model

There are two states of the world, $\Theta = \{L, H\}$. The state θ is drawn at the start and does not change. There is commonly held prior belief $q \in (0, 1)$ on the state being $\theta = L$.

A principal, which I refer to as the platform, seeks to take an action depending on the state. It has finite action set A and state-dependent utility function $u : A \times \Theta \to \mathbb{R}$.

Reviewers have information about the state. There are N reviewers, indexed by $i \in \{1, ..., N\}$. Each reviewer has a signal $S_i \in \mathbb{R}$, drawn from measure space $(\Omega, \mathcal{F}, P)$. Conditional on θ the S_i are independent, with densities given by f_θ. Signals are informative, so that $f_L \neq f_H$. In addition to her signal, reviewer i has a willingness to review $W_i \in \mathbb{R}$. The W_i are independent from each other and the S_i.

The principal chooses a review system, which is a mapping from signals to reviews, $r : \mathbb{R} \to \mathbb{R}$ (with r measurable). If the platform chooses r, and a reviewer with signal s_i submits her signal, the platform observes $r(s_i)$. Let $R_r = r(\mathbb{R})$ denote the set of reviews possible under r.

To submit a review, a reviewer bears a cost, that can be interpreted as either a cognitive or temporal cost. This is given as $c(r) = C(|R_r|)$ for some increasing function $C : \mathbb{N} \cup \{\infty\} \to \mathbb{R}$. This captures the intuition that it is "easier" to submit reviews when there are fewer choices.[2]

[1] Frick, Iijima, and Ishii (2023) and Frick, Iijima, and Ishii (2024a) apply similar techniques to different settings.

[2] Much recent work has shown that individuals have preferences against complexity. See, for instance, Oprea (2020). In some cases, complexity has been conceptualized in a way dependent on

After the platform makes its choice of review system, reviewers' private signals and willingness to review are drawn, and reviewers then decide whether to submit their review. Reviewer i receives intrinsic utility $w_i - c(r)$ from reviewing, and 0 utility from not. It follows that reviewers will use a simple threshold rule: if $w_i \geq c(r)$ reviewer i will review, and if $w_i < c(r)$ she will not. Let $p_r := P(W_i \geq c(r))$ be the ex-ante probability that a reviewer submits a review r.

After the N reviewers make their review decisions, the platform observes the submitted reviews, updates its beliefs, and then takes an action to maximize its expected utility. Explicitly, let $I \subseteq \{1, ..., N\}$ be the set of reviewers who choose to submit a review. The platform observes $\mathcal{S}(I) = (\tilde{r}(s_1), ..., \tilde{r}(s_N))$, which is the vector of *submitted* reviews:

$$\tilde{r}(s_i) = \begin{cases} r(s_i) & \text{if} \quad i \in I, \\ \emptyset & \text{if} \quad i \notin I. \end{cases}$$

After observing $\mathcal{S}(I)$, the platform updates its belief that the state is L. Let $\pi(\mathcal{S}) = \pi(L|\mathcal{S}(I))$ denote the posterior belief on the state being L after observing reviews $\mathcal{S}$. Here, and below, I drop the reliance on I for notational clarity.

After updating beliefs, the platform then chooses an action to maximize its expected utility given beliefs $\pi(\mathcal{S})$:

$$u^*(\mathcal{S}) := \max_{a \in A} \{\pi(\mathcal{S})u(a, L) + (1 - \pi(\mathcal{S}))u(a, H).\}$$

Since $\mathcal{S}$ is a random vector, $u^*(\mathcal{S})$ is stochastic. Define $u^*(r, N) := \mathbb{E}[u^*(\mathcal{S})]$, where this expectation is taken over the S_i and W_i. The objective of the platform is to maximize $u^*(r, N)$:

$$\max_r u^*(r, N).$$

the number of objects under consideration (e.g., Puri (2022)). Recently, Wang and Li (2025) studied the impact of increased cognitive load on user engagement, and found that decreases in complexity lead to more user engagement.

2.1 Discussion of Model

This model flexibly captures situations where an agent must use information elicited from others in order to make informed decisions but the agent cannot force or otherwise incentivize those with information to share their signals.

Importantly, I do not take a stance on the goal of the platform. In some settings, the platform's goal is to share the information that it collects with other agents (e.g., consumers of a product that is of unknown quality). The model naturally nests this setting, but does not assume that the platform's goals are aligned with future consumers. In this way, the model sheds insight on a variety of settings, from online retailers to general information dissemination platforms.

This highlights a major distinction between the platform in my model and previous models of review systems. For instance, the platform's goal in designing a rating/recommendation system in Acemoglu et al. (2022) and Che and Hörner (2018) is to maximize the expected utility of consumers. In my work, the platform's incentives may not be aligned directly with consumers. This difference is due to the primary focus in those works being how platforms can incentivize early consumers to explore a product of unknown quality, whereas I abstract from the problem of experimentation. For this reason, I view the present work as complimentary with the aforementioned studies. Previous works provide guidance on how a platform should design the information that it provides to early consumers to incentivize experimentation, while my study suggests how the platform should collect information in order to further its own goals. In practice, a platform can simultaneously maximize the information that it collects for itself and consider what information is optimal to show to consumers; the two objectives are not incompatible.

One of the main restrictions in the set-up above is the assumption that, given a review system r, the reporting rate p_r is uniform across the population of reviewers. In practice, this is unrealistic: it is natural that reviewers with more extreme signals are more likely to submit reviews.[3] To account for this, I extend the model to allow for non-uniform reporting rates in Section 5.1. The main insights of the baseline model extend to that setting, and so I begin with a discussion of the uniform reporting rates for clarity of exposition.

[3] See, for instance Lafky (2014).

As a final note of discussion on the model, I assume throughout that there are no processing costs for the platform to interpret the submitted reviews. One of the practical benefits to using simple rating schemes is that they are easier to digest and summarize than full-length reviews. My model abstracts from this, and assumes that there is no cost to interpreting the reviews. Closely tied to this is the possibility of miscommunication that arises from complicated reviews (e.g., free-text). I assume that when a reviewer submits a review, it is reported perfectly. Abstracting from these forces suggests that my model overstates the amount of information loss that stems from using coarsened signals, since removing these forces makes full-signal reports more accurate.

3 Quantifying Information Loss

In this section, I quantify the trade-off between informational content and frequency of report for different review systems by applying large-deviations techniques. I first show how a platform trades off between quality (appropriately defined) and frequency in review systems. I then characterize how much information a finite review system loses relative to the full review system when reviewers' signals are imprecise in order to characterize the optimal review system.

3.1 Trading-off between Quality and Frequency in Reviews

Fix a review system r. Denote the measure of reviews under r conditional on state L as γ_L^r. That is, for any measurable subset of possible reviews $B \subseteq R_r$ (where measurability is inherited from the Borel σ-algebra),

$$\gamma_L^r(B) := \mathbb{P}_L(r^{-1}(B)) = \int_{\{s:r(s)\in B\}} f_L(s)ds.$$

Similarly, let γ_H^r denote the measure corresponding to the distribution of reviews in state H. These measures reflect the distribution of reviews r *conditional* on reviews being submitted.

The performance of a review system depends on the rate at which the state is revealed under the review system. If γ_H^r and γ_L^r are very different, then learning will be fast. Consider, for a fixed review $r_i \in R_r$, the log-likelihood ratio of observing r_i

in states L and H:

$$\log\left(\frac{d\gamma_L^r(r_i)}{d\gamma_H^r(r_i)}\right). \tag{1}$$

In state H, learning will be fast if (1) is small, on average. The measure of difference between the distributions that determines this effect is the moment-generating function of (1) in state H, which is known as the *Hellinger transform* between the two distributions. The smaller the value of the Hellinger transform, the more likely the platform will think that the state is H when the true state is H after observing reviews.

Definition 1. The *learning efficiency* of review system r is the following measure of distance between γ_H^r, γ_L^r:

$$\nu(r) := 1 - \min_{\lambda\in[0,1]} \int_{R_r} \left(\frac{d\gamma_L^r}{d\gamma_H^r}\right)^{\lambda} d\gamma_H^r \in [0,1].$$

The learning efficiency is a simple transformation of the minimal value of the Hellinger transform. The less the two distributions agree, the larger learning efficiency becomes. At one extreme, when $d\gamma_L^r > 0 \iff d\gamma_H^r = 0$, then $\nu(r) = 1$, so that learning occurs immediately. At the other extreme, when reviews are entirely uninformative, $\gamma_L^r = \gamma_H^r$, $\nu(r) = 0$, and learning never occurs.

To see why the learning efficiency $\nu(r)$ is important, consider first the case that all reviews are submitted ($p_r = 1$). Then, beliefs after observing N reviews are determined by the behaviour of the sum of the log-likelihood ratios. In particular, for large N, the probability that the platform takes an action that is not optimal given the state is determined by the tail behaviour of the distribution of log-likelihood ratios. This behaviour is captured by $\nu(r)$, so that the learning efficiency captures the rate at which the platform learns the state. As a result, *regardless* of the platform's utility function u, $\nu(r)$ determines how quickly its expected utility converges its the full-information utility. Hence, if $\nu(r) > \nu(r')$, for large N the platform will have higher expected utility under r than under r'.[4]

[4]This discussion summarizes a known result in the literature (see, for instance, Chernoff (1952) and Torgersen (1981)).

When reviews are not always submitted ($p_r < 1$), the rate at which the platform learns the state is not $\nu(r)$. Intuitively, if reviews are submitted half of the time, the platform will learn half as quickly. The main result of this section is that this intuition is correct: regardless of the decision problem, for large N, $p_r\nu(r)$ determines the utility of the platform. This suggests that while the platform must trade-off between the rate at which reviewers are submitted and the quality of the reviews, the trade-off is separable.

Theorem 1. *Fix two review systems r, r' with corresponding rates of review $p_r, p_{r'}$ such that $p_r\nu(r) > p_{r'}\nu(r')$. For any finite action set A and utility function u, there exists an $\overline{N}$ such that for all $N \geq \overline{N}$, such that $u^*(r, N) > u^*(r', N)$.*

Theorem 1 provides a straightforward way for a platform to trade-off between frequency of reviews and their quality. Since $p_r\nu(r)$ is independent of the decision problem of the platform, the platform need not tailor its review design on the specifics of each problem. This is a particularly important consideration for large organizations where designers of the review system may be different from those using the information.

The proof of Theorem 1 leverages existing results in the large deviations literature which formalize the intuition above for the comparison when $p_r = 1$. In order to apply those results, I construct a statistical experiment with full reporting whose learning efficiency is $p_r\nu(r)$ and show that this experiment is equivalent to the review system with stochastic reporting.

In general, the review frequencies p_r are easy to empirically compute for platforms: an A/B test can quickly determine relative reporting rates for two different review systems. This suggests that it is beneficial to focus directly on the computation and comparison of the $\nu(r)$. An equivalent condition to that of Theorem 1 is, if

$$\frac{\nu(r')}{\nu(r)} < \frac{p_r}{p_{r'}}, \tag{2}$$

then there exists an $\overline{N}$ such that for all $N \geq \overline{N}$, such that $u^*(r, N) > u^*(r', N)$. The value $\nu(r')/\nu(r)$ determines the benchmark (in terms of relative frequency of reporting) that the platform uses in choosing between the two review systems. Consider the full review $r_f(s) = s$. By setting $r' = r_f$, the value (2) becomes a measure of how much information is lost in review r relative to the full review r_f.

Definition 2. The *learning loss* of a review function r is $\kappa(r) := \frac{\nu(r_f)}{\nu(r)} \in [1, \infty)$.

The learning loss $\kappa(r)$ of a review system r is how much more frequently it must be submitted relative to a full review for the platform to prefer eliciting reviews of type r to full reviews. For instance, if $\kappa(r) = 2$, then r must be submitted twice as frequently as a full review for the platform to prefer r when the number of reviewers is large.[5] The next section characterizes learning loss in a setting natural to online reviews.

Remark 1. It is important for the separability of $\nu(r)$ and p_r that the reporting rate p_r is uniform across the population of reviewers. If this is not the case (e.g., individuals with more extreme signals are more likely to leave a review), then the two are no longer completely separable. However, insights from the general model still apply in that setting. In Section 5.1 I extend the model to allow an individual reviewer's reporting rate to depend on their realized signal.

Remark 2. That the p_r represent the stochastic rate at which signals are reported is important. An alternative to this approach is to ask how many signals n_r from r "equals" $n_{r'}$ signals from r'. I discuss this alternative and its relationship to my specification in Section 5.2.

3.2 Finite Reviews and Scarce Information

When individual signals are not very informative, learning from any review system will be slow. In this setting, optimizing the rate of learning is even more important, since small sample effects will have a minor impact. This section studies the case of scarce information, and classifies $\kappa(r)$ in this setting.

In order to parameterize informativeness, I now assume that a reviewer's signal is given by $S_i = u_\theta + \epsilon_i$, where

$$u_\theta = \begin{cases} -\mu & \text{if} \quad \theta = L \\ \mu & \text{if} \quad \theta = H \end{cases}$$

[5]To compare two different review systems r and r', the platform simply uses $\kappa(r)/\kappa(r')$.

The parameter μ determines the separation between the two distributions: the larger the value of μ, the easier it is to distinguish the state. For large μ the two distributions overlap little, and for small μ the two are almost indistinguishable. The ϵ_i reflect the noise in the reviewers' experience of the good: different reviewers will experience the same product differently. For this reason I call ϵ_i reviewer i's taste idiosyncrasy.

Assumption 1. The idiosyncratic taste noise ϵ_i are i.i.d., with continuous, full-support, log-concave density given by f.

In this setting, the densities of signals in state L and H are given by

$$f_L(s) = f(s + \mu) \quad \text{and} \quad f_H(s) = f(s - \mu).$$

The two primitives of the model are f and μ. Given a review system r, write the learning efficiency of r when the separation is μ as $\nu_f(\mu; r)$. Similarly, write the learning loss of r as $\kappa_f(\mu; r)$. When it is clear from the context, I drop the dependence on f.

Threshold Rules: In this section I focus on the comparison between finite review reviews and the full review. In general, a finite review ($|R_r| = k$, for some $k \in \mathbb{N}$), is a (measurable) partition of $\mathbb{R}$ into k sections. In practice, review systems are monotone in that a more negative experience leads to a more negative rating. For arbitrary f, these will not in general be optimal. However, log-concavity of f guarantees that "threshold" review systems are optimal.

Explicitly, a review system r is called a k-threshold rule if there exist $\tau_0 < \tau_1 < ... < \tau_k$ with $\tau_0 = -\infty$ and $\tau_k = \infty$ such that

$$r(s) = \sum_{i=1}^{k} r_i \mathbb{1}\left\{s \in (\tau_{i-1}, \tau_i]\right\}, \text{ for some } r_i \in \mathbb{R}$$

In this case, I write $\tau^k = (\tau_1, ..., \tau_{k-1})$,[6] and write r_{τ^k} for the review system defined by thresholds τ^k. Optimality of threshold rules follows from (i) the log-concavity of f guaranteeing that the more negative the signal, the larger the belief that state is

[6]Since $\tau_0 = -\infty$ and $\tau_k = \infty$ are given.

$\theta = L$ (i.e., the Monotone Likelihood Ratio Property), and (ii) it is always being in the platform's interest to group together signals that are more informative of one state than the other.

Lemma 1. *Suppose that f is log-concave. Let r be a review system with $|R_r| = k$. For any μ, there exists a k-threshold rule r_{τ^k} such that $\nu(\mu; r_{\tau^k}) \geq \nu(\mu; r)$.*

Lemma 1 greatly simplifies the search of the platform: to implement an optimal review system with k reviews, it optimizes over the $k-1$ thresholds τ_i.

Scarce Information: In practice, individual reviews are largely uninformative, due to the large impact of taste idiosyncrasies. This corresponds to $\mu \approx 0$.[7] In order to understand this setting better, I study

$$\kappa(0; r_{\tau^k}) := \lim_{\mu \to 0} \kappa(\mu; r_{\tau^k}) = \lim_{\mu \to 0} \frac{\nu(\mu; r_f)}{\nu(\mu; r_{\tau^k})}. \tag{3}$$

Although $\kappa(0; r_{\tau^k})$ is not well-defined (as $\nu(0; r) = 0$ for all r), I abuse notation and write it to represent the limit in (3).[8] This limit represents the relative rates of reporting that make the platform prefer eliciting reviews r_{τ^k} to eliciting full reviews when information is noisy. If the limit (3) is large, then when information is scarce, a binary review system with thresholds τ^k performs poorly relative to full reviews. On the other hand, if (3) is small, then even when reviewers are slightly more likely to leave a binary review, the platform would prefer eliciting coarse information.

The main result of this section is the explicit characterization of $\kappa(0; r_{\tau^k})$. Before I introduce the result, some intuition is useful. When $\mu = 0$, $\nu(\mu; r_f) = 0$, since there is no difference in the distribution of signals in the two states. When the separation μ increases from 0, on regions where f is nearly constant, signals will continue to be uninformative. However, on regions where f is very curved, even a small shift in μ makes signals in that region very informative, because the posteriors that these signals induce will shift greatly. A larger curvature of f is then associated with a larger value of (3). The measure of curvature that captures this effect is the *Fisher*

[7] An alternative approach is to take the variance of f to be large. The two approaches are equivalent (see Remark 3 at the end of this section).

[8] The optimal $(\tau^k)^*$ will depend on μ. However, it is sufficient to set $(\tau^k)^* = \lim_{\mu \to 0} (\tau^k)^*(\mu)$ as $\lim_{\mu \to 0} \frac{\nu(\mu; (\tau^k)^*(\mu))}{\nu(\mu; (\tau^k)^*)} = 1$.

Information of f (with respect to μ):

$$I_f := \mathbb{E}\left[\left(\frac{f'(s)}{f(s)}\right)^2\right].$$

A similar logic determines the rate at which $\nu(\mu; r_{\tau^k})$ increases as the separation μ increases from zero. Since the review is discrete, the effect of a marginal change in μ will only impact the reviews at the thresholds $\tau_1, ..., \tau_{k-1}$. Consider that for a single review $r_i \in R_{r_{\tau^k}}$, a marginal increase in the separation marginally increases the probability that r_i is reported in state L by $f(\tau_i) - f(\tau_{i-1})$. The probability that r_i is reported in state H decreases by the same amount. The larger the absolute value of this difference, the more informative review r_i becomes as μ increases. This effect is magnified if r_i is uncommon because the difference is more noticeable.

The following assumption is a regularity assumption on f, guaranteeing that the Fisher information is well-defined.[9]

Assumption 2. The density of idiosyncratic taste shocks f is absolutely continuous almost everywhere, as is its derivative.

These effects come together in the following major result of this section:[10]

Theorem 2. *Suppose that f that satisfies Assumptions 1 and 2. Let F denote the cumulative distribution function of f. Then, learning loss with scarce information is given by*

$$\kappa(0; r_{\tau^k}) = I_f \Bigg/ \left(\sum_{i=1}^{k} \frac{(f(\tau_i) - f(\tau_{i-1}))^2}{F(\tau_i) - F(\tau_{i-1})}\right) \in [1, \infty).$$

Theorem 2 gives a straightforward way to determine the degree of information loss for different review functions. It is proved by showing independently the rate at which $\nu(\mu; r_f)$ and $\nu(\mu; r_{\tau^k})$ go to zero as a function of the separation of the distributions.

An important corollary of Theorem 2 is an explicit characterization of the optimal finite review system with $|R_r| = k$.

[9]A slightly weaker sufficient condition is that f is twice continuously differentiable almost everywhere, with the first and second derivative of $f(x+\mu)^\lambda f(x-\mu)^{1-\lambda}$ with respect to μ being integrable, for μ in a neighbourhood of 0.

[10]With the notational convention that $f(\infty) = f(-\infty) = 0$, and $F(\infty) = 1, F(-\infty) = 0$.

Corollary 1. *The maximizer of*

$$\sum_{i=1}^{k} \frac{\left(f(\tau_i) - f(\tau_{i-1})\right)^2}{F(\tau_i) - F(\tau_{i-1})}$$

is the optimal k-threshold rule r_{τ^k} under scarce information.

Corollary 1 provides a straightforward objective for the platform to maximize. It also shows that the distribution of taste shocks in the population has important implications for the design of review systems. In order to characterize the optimal review system across all possible review systems, the platform compares the optimal reviews with k thresholds, for all k. Depending on how much complexity impacts reviews, it may be optimal to choose very coarse systems in some settings and less coarse systems in others.

Remark 3. An alternative to parameterizing the distribution of signals by the separation μ would be to vary the variance:

$$f_L(s;\sigma) \propto f\left(\frac{s+\mu}{\sigma}\right) \quad \text{and} \quad f_H(s;\sigma) \propto f\left(\frac{s-\mu}{\sigma}\right).$$

Scarce information in this setting corresponds to taking $\sigma \to \infty$. The choice between the separation μ and variance σ parameterization is entirely stylistic: a parameterization of (μ, σ) yields the same information as $(\frac{\mu}{\sigma}, 1)$. The drawback of varying σ is that the notation regarding an optimal k-threshold rule becomes more cumbersome. Instead of having a well-defined limit τ_i, convergence is in terms of $\frac{\tau_i(\sigma)}{\sigma}$. Apart from the notation, the two are the same.

4 Application: Binary Reviews and Heterogeneity

In this section, I use the characterization from Theorem 2 to show how optimal review systems vary with the distribution of reviewers' taste idiosyncrasies. An important economic characteristic for information elicitation is the homogeneity of the population with information. Preferences over products like movies (or other forms of art) are dispersed because individual taste matters to a large degree. However, for simple products like toasters, preferences are much less dispersed, since experiences rely on simple characteristics.[11] I use this section to show that heterogeneity has important implications for review systems. I focus primarily on the case of binary reviews for ease of exposition.

In order to understand how taste heterogeneity affects review systems, I restrict attention to a one-parameter family of distributions where the parameter can be interpreted as the degree of homogeneity within the population of reviewers:

$$f_\alpha(s) \propto \exp(-|s|^\alpha). \tag{4}$$

Large tails correspond to dispersion in tastes, and hence heterogeneous idiosyncratic tastes. The parameter α in (4) captures this tail behaviour. The larger the value of α, the smaller are tails, and so, the smaller the dispersion of taste shocks. Henceforth, I refer to α as homogeneity.

Remark 4. For $\alpha \geq 1$, $f_\alpha(s)$ is log-concave, and so Lemma 1 implies that a threshold rule is optimal. This family is sometimes referred to as the Generalized Normal Distributions and incorporates three well-known distributions as special cases: f_1 is the Laplace distribution (large heterogeneity); f_2 is the Normal distribution (moderate homogeneity); and as $\alpha \to \infty$, f_α converges almost everywhere to the uniform distribution on $(-1, 1)$ (perfect homogeneity). Since variance does not affect learning loss with scarce information, the scale parameter is normalized to 1.

Studying this class of distributions allows me to tractably show how heterogeneity impacts information loss and the optimal choice of threshold for binary review systems. To motivate why properties of the optimal threshold are important, I first show what happens when the platform uses a naive threshold: asking reviewers if their experience was positive or negative. This review corresponds to r_0, the review system with the symmetric threshold of 0.

When the population is very heterogeneous, this threshold leads to small learning loss. However, as homogeneity grows, this does not continue: the naive threshold

[11] One possible way to distinguish between goods with low and high taste homogeneity would be the classification of "search" and "experience" goods used in the empirical literature. First introduced in Nelson (1970), search goods are those goods whose attributes are well-defined and easily found, whereas experience goods are those which must be experienced before an opinion can be formed (see also Magnani (2020) for a discussion). Search goods, because of their well-defined attributes, are likely to be more homogeneous ("does the good perform as it should?"), while experience goods are likely to be more heterogeneous (since individual experience important).

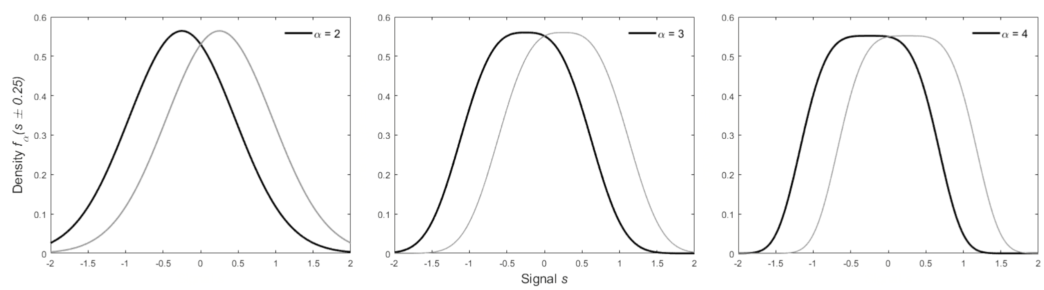
leads to unbounded loss in information.

Lemma 2. *Consider learning loss under the naive threshold 0, $\kappa_{f_\alpha}(0; r_0)$:*

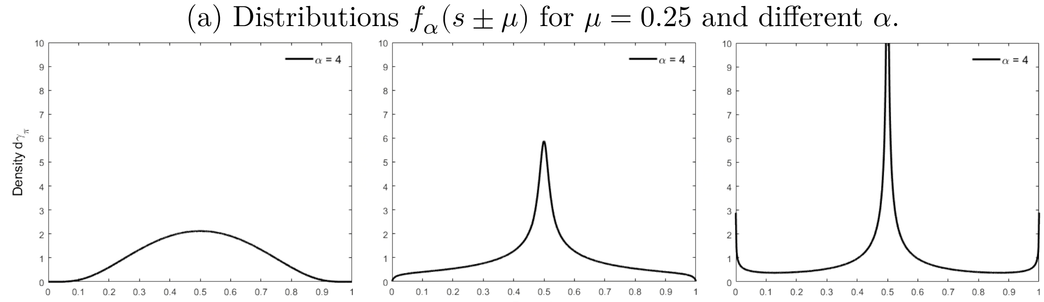
(i) $\kappa_{f_1}(0; r_0) = 1$*; and*

(ii) $\kappa_{f_\alpha}(0; r_0)$ *is increasing in* $\alpha \geq 1$ *with* $\lim_{\alpha \to \infty} \kappa_{f_\alpha}(0; r_0) = \infty$.

Lemma 2 shows that under the naive binary review, information loss is increasing in the degree of homogeneity of the population. The intuition is as follows. When idiosyncratic tastes are less dispersed, while many signals are very uninformative, there are also more signals that are very informative (i.e., some reviewers are very representative). See Figure 1, which shows this visually. As homogeneity α increases, simultaneously more signals are very uninformative and more signals are very informative. In posterior space, this is reflected in a larger spread of the induced posteriors.[12] Using the symmetric review r_0 mixes very informative signals with very uninformative signals, limiting the information contained in either review.



(a) Distributions $f_\alpha(s \pm \mu)$ for $\mu = 0.25$ and different α.



(b) Distributions $f_\alpha(s \pm \mu)$ for $\mu = 0.25$ and different α.

Figure 1: Different distributions of $f_\alpha(s \pm \mu)$ and corresponding ex-ante distributions of posteriors (for uniform prior) for $\mu = 0.25$ and different values of α.

[12] Appendix B explores this relationship in more detail.

This discussion also highlights the central tension in choosing an optimal threshold: extracting more information from one review while extracting less from the other. When α is small, there is not sufficient benefit to justify using an asymmetric review. However, when α is large, because there are some very informative signals, enough information can be extracted from the informative review to outweigh the cost of the other review being uninformative: the optimal binary review is asymmetric. Importantly, the optimal threshold limits learning loss under the binary review: the optimal review bounds learning loss by 3.25. That is, if reviewers are 3.25 times as likely to submit a binary review than their full signal, the optimal binary system is preferred, regardless of the degree of homogeneity. This result, together with Lemma 2, highlights the importance of the analysis: the choice of the threshold has extremely important implications for the performance of binary reviews.

In what follows, let $\tau^*(\alpha)$ be the nonnegative value that maximizes[13]

$$\frac{f_\alpha^2(\tau)}{F_\alpha(\tau)} + \frac{f_\alpha^2(\tau)}{(1 - F_\alpha(\tau))}. \tag{5}$$

Proposition 1. *For $\alpha \geq 1$, there is a unique maximizer $\tau^*(\alpha)$ to (5) on $[0, \infty)$. It has the following properties:*

(i) If $\alpha \leq 2$, $\tau^*(\alpha) = 0$, while if $\alpha > 2$, $\tau^*(\alpha) \neq 0$;

(ii) $\lim_{\alpha \to \infty} \tau^*(\alpha) = 1$; and

(iii) $\kappa_{f_\alpha}(0; r_{\tau^*(\alpha)}) < 3.25$ for all α.

The normal distribution corresponds to the degree of homogeneity at which the optimal review ceases to be symmetric. When the degree of homogeneity is large the platform optimally asks individuals either (i) whether they had a very good experience or an experience that was not exceptional (positive τ^*), or (ii) whether they had a very bad experience or an experience that was not horrendous (negative τ^*).

Figure 2 plots comparative statics for $\tau^*(\alpha)$: both $\tau^*(\alpha)$ and $F_\alpha(\tau^*(\alpha))$ are increasing in α. Importantly, the probability of the more common review increases in

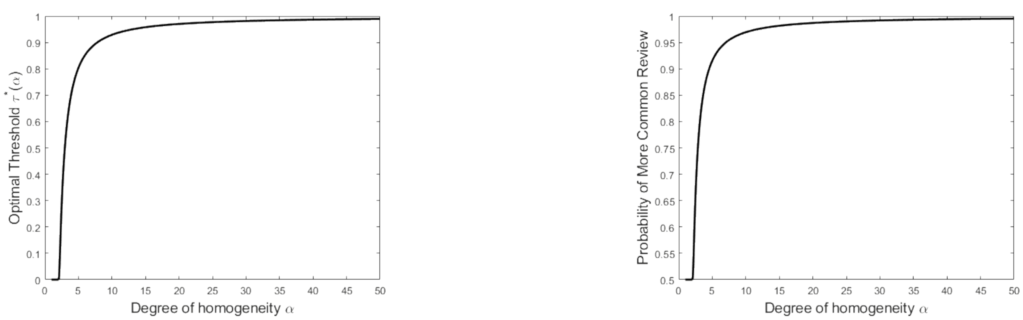
[13]Since f_α is symmetric, there will be also a nonpositive maximizer $-\tau^*(\alpha)$. In the statement of Proposition 1, everything is written in terms of the nonnegative optimal threshold for clarity of exposition. Comparable statements for the nonpositive optimal threshold hold. Section 4.1 shows that introducing (arbitrarily small) asymmetry in f_α breaks this multiplicity.

α, *regardless of the state.* Consider $r_{\tau^*(\alpha)}$, the optimal binary review with a positive threshold. For large α, even when $\theta = H$, the majority share of reviews will be negative. This share increases to 1 as $\alpha \to \infty$, showing that for large degrees of homogeneity, the optimal review system appears to degenerate. This suggests that even review systems where the vast majority of reviewers leave the same review may not be sub-optimal. These results are summarized in Proposition 2.

Proposition 2. *Consider the non-negative optimal threshold* $\tau^*(\alpha)$. *The following are true:*

(i) The optimal threshold is increasing in the homogeneity of the population: $\tau^*(\alpha)$ *is strictly increasing in* α *for* $\alpha \geq 2$*; and*

(ii) Optimal asymmetry is increasing in the homogeneity of the population: $F_\alpha(\tau^*(\alpha))$ *is strictly increasing in* α *for* $\alpha \geq 2$.



(a) As homogeneity increases, the optimal review becomes more asymmetric.

(b) As homogeneity increases, one of the reviews becomes extremely rare.

Figure 2: Properties of the optimal binary review system, for different degrees of homogeneity α.

4.1 Asymmetric Heterogeneity

This section shows that the multiplicity of optimal reviews disappears when any asymmetry in the distribution of taste shocks is introduced. In particular, I introduce asymmetry in the degree of homogeneity for positive and negative idiosyncratic tastes. For any nonzero difference in homogeneity on the two sides of the distribution, the platform optimally chooses a threshold on the side of the distribution that is more homogeneous.

Apart from breaking multiplicity, this is an important setting because asymmetric homogeneity is common. In hiring a taxi or ride-share service, the qualities that lead to a very bad experience are likely to not differ greatly across individuals: no customer wants to be late or get into an accident. On the other hand, the qualities that lead to a very good experience may differ across individuals: some (not all) people like having a long discussion with their driver; some (not all) like music to be played.[14] When negative experiences are more homogeneous the platform optimally uses negative reviews only for very bad experiences: the optimal threshold is negative. This result explains why many review systems are positively skewed: most reviews on Airbnb are 5-stars, as are those on Uber (see Hu, Zhang, and Pavlou (2009)).

Let α^* be the value for which $f_\alpha(0)$ is maximized.[15]

Proposition 3. *Let*

$$f_{\alpha_L,\alpha_H}(s) \propto \mathbb{1}\{s \geq 0\}e^{-|s|^{\alpha_H}} + \mathbb{1}\{s < 0\}e^{-|s|^{\alpha_L}}$$

with $\alpha^* \leq \alpha_H, \alpha_L$. If $\alpha_L > \alpha_H$, then, the optimal threshold $\tau^*(\alpha_L, \alpha_H)$ is negative. If $\alpha_H > \alpha_L$, then the optimal threshold is positive.

Figure 3 highlights the intuition behind Proposition 3. Consider a positive threshold (dotted line). At the negative threshold that yields the same value of the density (dashed line), it is easily seen that there is less mass to the left of that threshold than there is to the right of this positive candidate. This logic applies to any possible positive candidate, so that the optimal threshold must be negative.[16]

Proposition 3 justifies why many platforms exhibit many more positive reviews than negative reviews. If people agree on negative experiences more than they do positive experiences, for optimal learning, the platform *should* isolate extremely negative experiences and thus choose a system in which a large portion of the population leaves a positive review, even if the product is of low quality.

[14] As another example, consider again search and experience goods. For most search goods, increased negative homogeneity is expected: most reviewers will be in agreement about whether a good doesn't meet expectations, but there may be disagreement about how it over-performs.

[15] It is easy to verify that $\alpha^* \approx 2.166$.

[16] A similar logic applies if, instead of increasing the degree of positive heterogeneity, the variance of signals for positive taste shocks. In that case, too, the optimal threshold would necessarily be negative.

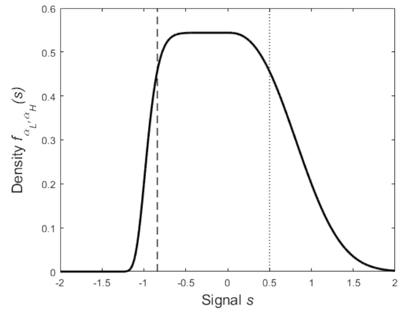
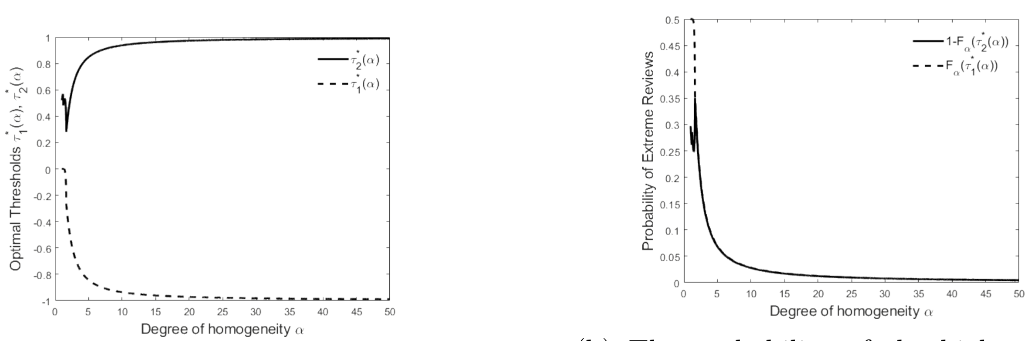


Figure 3: Asymmetric distribution of taste shocks f_{α_L,α_H}, for $\alpha_L = 10, \alpha_H = 2.5$.

4.2 Heterogeneity with Multiple Bins

Similar qualitative results apply to the finite review systems with k bins, where $k > 2$. Figure 4 exhibits the optimal three-bin system for different degrees of homogeneity α. In particular, learning loss is increasing in the degree of homogeneity, and the optimal asymmetry of the system is also increasing in the degree of homogeneity. An important difference here over the binary system is that *both* sides of the distribution can be isolated. As homogeneity grows, it is increasingly important to "throw out" large portions of the population. The third bin does not provide any information to the platform (since it is equally likely in each state). Its presence is important, however, because it allows the platform to skim uninformative reviews in order to isolate very informative positive and negative signals.



(a) The optimal thresholds become more extreme as homogeneity grows.

(b) The probability of the highest and lowest review shrinks as homogeneity grows.

Figure 4: Properties of the optimal three-bin review system, for different degrees of homogeneity α.

5 Extensions and Discussion

5.1 Non-Uniform Reporting Rates

I assume above that given a review system r, the rate of reporting p_r is uniform across the population of reviewers. In practice a reviewer's experience is likely to affect the probability that they submit a review. It is reasonable that a reviewer with an extreme review is more likely to leave a review, and in some contexts reviewers may report positive or negative experiences at different rates.[17] In this section I show how to incorporate non-uniform reporting rates into the framework studied above. While the qualitative insights from previous sections remain, several new forces emerge.

Suppose that a reviewer with signal s facing review r has a probability of leaving a review given by $p(r, s)$, dependent on both the review system *and* the reviewer's signal. As before, it is the difference in review measures across the two states that determines the platform's learning rate. In this setting, however, it is not possible to separate the frequency of reviews from the underlying signal distribution. Explicitly, define the measure $\gamma_L^{(p,r)}$ of reviews in state L (implicitly dependent on μ) as

$$\gamma_L^{(p,r)}(B) := \int_{\{s:r(s)\in B\}} p(r, s) f(s + \mu) ds.$$

Similarly define the measure $\gamma_H^{(p,r)}$ of reviews in state H. As before, these measures are used to determine the rate at which the platform learns the state. Define $\rho(\mu; r, p)$ as the analog of $p_r \nu(\mu; r)$:

$$\rho(\mu; r, p) := 1 - \min_{\lambda\in[0,1]} \left[\int_{R_r} \left(\frac{d\gamma_L^{(p,r)}}{d\gamma_H^{(p,r)}} \right)^{\lambda} d\gamma_H^{(p,r)} + \left(\gamma_L^{(1-p,r)}(\mathbb{R})\right)^{\lambda} \left(\gamma_H^{(1-p,r)}(\mathbb{R})\right)^{1-\lambda} \right].$$

There are two significant differences between ρ and $p_r \nu(r)$. First, $p(r, \cdot)$ varies with the signal realization, different signals receive different weight. Second, "null reviews" (reviews that are not submitted) can now be informative of the state. Null reviews are an additional signal: if reviewers are more likely to be submitted when $\theta = L$, a lack of reviews is an indication that $\theta = H$. Theorem 1 generalizes to this

[17] For instance, Lafky (2014) suggests that individuals are more likely to report extreme experiences than moderate experiences because they derive a benefit from informing others, and extreme experiences are more informative.

setting, where the $\rho(\mu; r, p)$ take the role of $p_r \nu(\mu; r)$.

Lemma 3. *Fix two review systems r, r' such that $\rho(\mu; r, p) > \rho(\mu; r', p)$. For any action set A and a utility function u, there exists an $\overline{N}$ such that for all $N \geq \overline{N}$, such that $u^*(r, N) > u^*(r', N)$.*

Lemma 3 shows that even with full flexibility in reporting rates a generic ordering over review systems exists. Without additional structure on the propensities to review p, however, little more can be said about the comparison. When arbitrary functions p are allowed, then the underlying distribution of signals that different review systems draw from differ, and anything can happen. In order to gain structure, I restrict attention to functions p that are separable between the impact of the review function and the signal realization.

Definition 3. The propensity to review function $p(r, s)$ is said to be *separable* if $p(r, s) = p_r q(s)$ for some functions $p : \mathbb{N} \cup \{\infty\} \to (0, 1]$ and $q : \mathbb{R} \to [0, 1]$.

This specification represents a two-stage procedure. The first stage reflects individuals who are willing to ever submit a review of type r (a fraction $p_r = p(|R_r|)$ of the population). In the second stage, these individuals make their decision based off of their signal realization: among these, only fraction $q(s)$ of those with signal s will decide to leave the review.[18] This reflects that even reviewers who sometimes leave reviews will not always leave them, and this choice may be influenced by their experience.[19]

To understand the new forces that enter when reporting rates are not uniform, I now develop an analogue of Theorem 2 in the case that propensities to review are separable. Before I present the result, additional notation is needed. Fix the distribution of taste heterogeneity f and the signal reporting rate q. Define the functional E as the integral of a function g with respect to the measure whose density is given by $q \cdot f$:

$$E(g) := \int_{-\infty}^{\infty} g(s) q(s) f(s) ds.$$

[18] A micro-foundation for this two-stage procedure is given in Appendix A.3.

[19] Importantly, q depends on the signal and *not* the taste shock ϵ_i of the individual. This latter case would simply reduce to the setting of the previous sections.

Note that E is not an expectation because in general $q \cdot f$ is not probability distribution (unless $q \equiv 1$ almost everywhere). Similarly, given a review function r, and a review $\tilde{r} \in R_r$, denote by

$$E(g; \tilde{r}) := \int_{-\infty}^{\infty} g(s) \mathbb{1}\{r(s) \in \tilde{r}\} q(s) f(s) ds$$

the integral with respect to the restricted measure. With this notation, Theorem 2 generalizes to the following result.

Theorem 3. *Suppose that f satisfies Assumptions 1 and 2 and $p(r,s) = p_r q(s)$ where q is continuous. Let $\ell_f := \frac{f'}{f}$ denote the linear score of f. Then learning loss for a finite review system r is given by*

$$\frac{p_r}{p_f} \cdot \lim_{\mu \to 0} \frac{\rho(\mu; p_f \cdot q)}{\rho(\mu; r, p_r \cdot q)} = \frac{E\left(\ell_f^2\right) + \frac{p_f E(\ell_f)^2}{1 - p_f E(1)}}{\sum_{\tilde{r} \in R_r} \frac{E(\ell_f; \tilde{r})^2}{E(1; \tilde{r})} + \frac{p_r E(\ell_f)^2}{1 - p_r E(1)}}. \tag{6}$$

The left-hand side of (6) is the generalization of κ from Section 4. In the case that $q \equiv 1$ and r is a threshold rule, Theorem 3 is a generalization of Theorem 2 for non-threshold finite reviews.[20] The first term of the numerator and the sum in the denominator are the direct generalizations of the numerator and denominator from Theorem 2.

The impact of null reviews is reflected in the second term of the numerator and denominator. Importantly, the information contained in null reviews is related to the rate at which a review is submitted. In Figure 5, the top two panels show the distribution of submitted reviews. As p_r increases, these distributions are scaled. However, this is not the case for null reviews, which are shown in the bottom two panels. As the fewer reviewers report ($p_r \to 0$), the mass of null reviews becomes more similar in the two states, since $1 - q(s)p_r$ approaches 1 uniformly (drowning out any asymmetry due to q). Hence, null reviews become less informative as p_r decreases. In particular, in the case when few people submit reviews (i.e., $p_r \approx 0$), the only difference that non-uniform reporting rates introduce is mechanical, since null reviews are not informative.

[20] Even in this setting, threshold rules will be optimal. However, the notation is such that it is easier to write (6) in full generality.

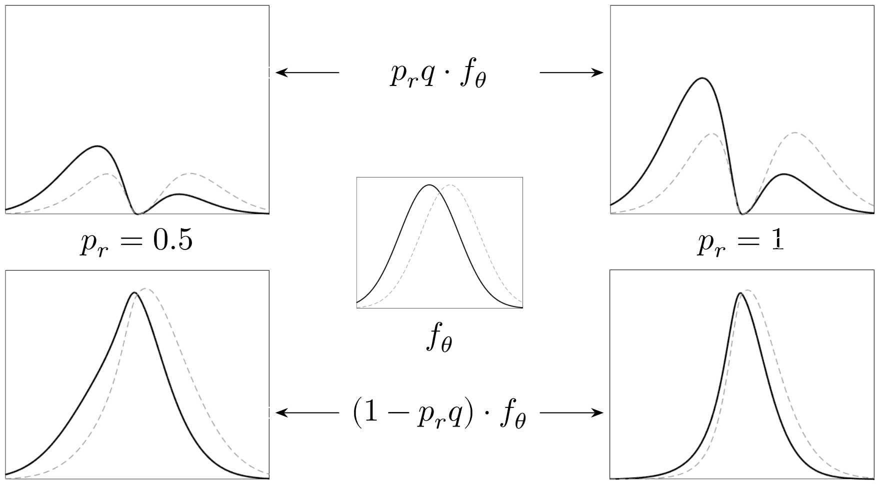


Figure 5: Distributions of reviews for $f_\theta(s)$ normal, with $\mu = 0.25$, and $q(s) = 1 - \frac{1}{1+2(x^2+4\mathbb{1}\{x\leq 0\}x^2)}$ and for $p_r = 0.5$ (left), 1 (right). The top panels show the distribution of submitted reviews, and the bottom panels show the distribution of null reviews.

5.2 Relationship between Production and Consumption of Information

As discussed above, the rates of reporting p (whether uniform or not) are taken from an ex-ante, and not an ex-post, perspective. An ex-post perspective would be to directly compare n_r reviews of type r with $n_{r'}$ reviews of type r'.[21] In practice, this latter decision falls to consumers who must decide between multiple types of reviews on a website. While the ex-ante and ex-post preferences over sources of information are related, they are not identical.

Consider two different review systems, r and r', with $\nu(r) \gg \nu(r')$. If $p_r \ll p_{r'}$, then the platform may be worried about not receiving any (or receiving very few) reviews of type r, and so might prefer r'. However, on average, there will still be many reviews of type r submitted, and so to a consumer deciding between Np_r reviews from system r and $Np_{r'}$ reviews from system r', this issue does not arise. As a result, the ex-post perspective leads to a stronger preference for finer reviews. However, when information is scarce, this difference disappears and the two orders agree. Appendix C formalizes the model and the differences, and shows these relationships.

[21]The ex-post perspective has been traditionally taken in the literature. For instance, see Chernoff (1952), Moscarini and Smith (2002), and Mu et al. (2021).

6 Discussion and Conclusion

My results suggest that if (i) the platform has complete control over how reviewers review, (ii) information is one-dimensional, and (iii) there are only two levels of product quality, then binary review systems perform very well. I briefly discuss these three points before concluding.

(i) Implementation: I abstract entirely from the implementation of review systems. In practice, controlling how reviewers leave reviews is likely difficult. Ubiquitous review systems, like 5-star reviews, are likely difficult to design because of reviewers' extensive experience with similar systems: reviewers are likely to default to their own, private, thresholds. Binary systems, because of their simplicity, are easier to carefully design (motivating my focus on them in Section 4). In practice, many binary reviews ask different questions—"did you like the product?", "would you buy the product again?", "would you recommend the product to a friend/colleague?"—all of which induce different thresholds. Exploring how to elicit information from reviewers who have past experience with different review systems is an interesting question for future work.

(ii) Multi-Dimensional Information: In practice, even simple products like toasters have different relevant dimensions of quality. While my results do not directly speak to this, they do suggest a naive approach of isolating different dimensions of product quality, and ask a binary question of quality for each of these dimensions. My results suggest that, because of the performance of binary systems in the one-dimensional case, even this naive approach will perform well: the same bound on relative rates of reporting will apply. An interesting question to ask is whether it is optimal to isolate dimensions in this way, or if it is better to combine information.

(iii) Multiple States: When there are more than two states of the world, my results can be extended in the natural way. First, the platform partitions states into groups that lead to different actions. Then, the rate of learning for the platform is determined by the two states across groups that are hardest to distinguish. In the case of binary action environments (fire/keep on, recommend/do not recommend, or settings where the goal is to flag if there is a problem), my analysis directly applies. This is because the platform can apply the results to the two states that are closest and lead to the two different actions, with the caveat that the optimal review system will depend on the decision problem (because the problem determines which two states are hardest to distinguish). In settings with a richer action space, more care must be taken because a review system that distinguishes two elements of the partition well might make it hard to distinguish a third. In this setting, binary review systems may perform poorly, but more complicated finite review systems may provide some robustness.

6.1 Conclusion

How to best elicit information from agents is an important economic question. My results apply more broadly than online review systems. Whenever a principal elicits information from agents it must consider the trade-off between quality and quantity of information. I show how a platform optimally designs a review system, and show that learning outcomes are sensitive to the chosen review system. Technically, I quantify, in a rate of learning sense, how much information is lost when signals are coarsened. I use this quantification to derive the trade-off between the quality of a review and how frequently it is reported and moreover characterize the optimal review system when reviewers' signals are noisy.

In order to understand the economic implications of the model, I apply the framework to the context of online reviews. The degree of taste homogeneity for the product being reviewed has important implications for how much information is lost when signals are coarsened and for the structure of optimal binary systems. As the population becomes more homogeneous, more information is lost. A symmetric binary system can lose an arbitrary amount of information relative to the full signal, but this loss can be greatly mitigated by optimally using an asymmetric system. My results explain why certain review systems are used in practice—in particular explaining the prevalence of positively-skewed systems and binary schemes—and highlight what forces are important for the performance of review systems.

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A Proofs

All proofs for results in the main text are proved in this appendix.

A.1 Proofs for Section 3

In this appendix I provide the proof of the major result from Section 3. Before I prove the main result, I first state the following result from Torgersen (1981):

Theorem (Torgersen 1981). *Suppose that reviews are always submitted (i.e., $p_r = 1$ for all review systems r). Fix two review systems r, r' with $\nu(r) > \nu(r')$ and a decision problem for the platform. There exists an $\overline{N}$ such that for all $N \geq \overline{N}$, such that $u^*(r, N) > u^*(r', N)$.*

Proof of Theorem 1. For ease of notation, write $\nu_r := \nu(r)$ to denote the learning efficiency of a review r. The pair (p_r, r) can be thought of as defining a statistical experiment over the two states of the world.

In order to apply the above theorem to this setting, we must translate (p_r, r) into a statistical experiment without stochastic reporting, and show that its learning efficiency is given by $p_r \nu_r$, where ν_r is the learning efficiency of the statistical experiment $(1, r)$.

Let $\gamma_L^{(p_r, r)}$ be the measure defined over $R_r \cup \{\emptyset\}$ representing the distribution of reviews in state L. That is,

$$\begin{aligned}\gamma_L^{(p_r,r)}(B) :=& p_r \cdot \int_{\{s: r(s) \in B\}} f_L(s) ds + (1 - p_r) \cdot \mathbb{1}\{\emptyset \in B\} \\ =& p_r \cdot \gamma_L^r(B \cap R_r) + (1 - p_r) \cdot \mathbb{1}\{\emptyset \in B\}.\end{aligned}$$

The only difference between this measure and the measure γ_L^r is the weight $1 - p_r$ placed on $\{\emptyset\}$, which represents a null review (a reviewer choosing to not submit a review). Similarly,

$$\gamma_H^{(p_r,r)}(B) := p_r \cdot \gamma_H^r(B \cap R_r) + (1 - p_r) \cdot \mathbb{1}\{\emptyset \in B\}.$$

Notice then, for a given value of λ, it is the case that

$$\int_{R_r \cup \{\emptyset\}} \left(d\gamma_L^{(p_r,r)}\right)^\lambda \left(d\gamma_H^{(p_r,r)}\right)^{1-\lambda} = \int_{R_r} (p_r \cdot d\gamma_L^r)^\lambda (p_r \cdot d\gamma_H^r)^{1-\lambda} + (1 - p_r)$$

$$=p_r \int_{R_r} \left(d\gamma_L^r\right)^\lambda \left(d\gamma_H^r\right)^{1-\lambda} + 1 - p_r.$$

Hence,

$$\min_{\lambda\in[0,1]} \int_{R_r\cup\{\emptyset\}} \left(d\gamma_L^{(p_r,r)}\right)^\lambda \left(d\gamma_H^{(p_r,r)}\right)^{1-\lambda} = 1 - p_r \left(1 - \min_{\lambda\in[0,1]} \int_{R_r} \left(d\gamma_L^r\right)^\lambda \left(d\gamma_H^r\right)^{1-\lambda}\right).$$

Concluding,

$$\nu_{(p_r,r)} = 1 - \min_{\lambda\in[0,1]} \int_{R_r\cup\{\emptyset\}} \left(d\gamma_L^{(p_r,r)}\right)^\lambda \left(d\gamma_H^{(p_r,r)}\right)^{1-\lambda} = p_r \cdot \nu_r,$$

which is what we set out to show. □

Proof of Lemma 1. This follows from the Monotone Likelihood Ratio Property of log-concave densities and Proposition 5, which is stated and proved in Appendix B. □

Proof of Theorem 2. The proof follows from an application to Theorem 3, in the case that $q \equiv 1$ and the review system r is a threshold system. To see this, observe that (with the notation of the proof of Theorem 3), if $\tilde{r}$ consists of those signals that fall between τ_{i-1} and τ_i, it is the case that

$$H'_+(0,\tilde{r})^2 = \left(\int_{\tau_{i-1}}^{\tau_i} p_r f'(s)ds\right)^2 = p_r^2 \left(f(\tau_i) - f(\tau_{i-1})\right)^2 \quad \text{and}$$
$$H_+(0,\tilde{r}) = \left(\int_{\tau_{i-1}}^{\tau_i} p_r f(s)ds\right) = p_r \left(F(\tau_i) - F(\tau_{i-1})\right).$$

The result follows from simply taking the ratio. □

A.2 Proofs for Section 4

In this Appendix I provide the proofs of the results in Section 4. To begin the analysis, some preliminary calculations are needed. Closely related to our family interest is the Gamma function.

Definition 4. Denote by $\Gamma(\alpha)$ the standard Gamma function, given by

$$\Gamma(\alpha) := \int_0^\infty x^{\alpha-1}e^{-x}dx.$$

Let $\Gamma(\alpha, t)$ denote the upper incomplete gamma function:

$$\Gamma(\alpha, t) := \int_t^\infty x^{\alpha-1} e^{-x} dx.$$

Finally, let $\gamma(\alpha, t)$ denote the lower incomplete gamma function

$$\gamma(\alpha, t) := \int_0^t x^{\alpha-1} e^{-x} dx = \Gamma(\alpha) - \Gamma(\alpha, t).$$

Lemma 4. *It is the case that*

$$f_\alpha(x) = \frac{\alpha}{2\Gamma(1/\alpha)} e^{-|x|^\alpha}.$$

Moreover, the Fisher information of f_α is given by

$$I_{f_\alpha}(0) = \frac{\alpha^2}{\Gamma(1/\alpha)} \Gamma\left(2 - \frac{1}{\alpha}\right).$$

Proof of Lemma 4. The first part of the statement is just ensuring that f_α is indeed a probability distribution.

$$\int_{-\infty}^\infty e^{-|x|^\alpha} dx = 2 \int_0^\infty e^{-x^\alpha} dx.$$

Make now the substitution $u = x^\alpha$. Then, $\frac{1}{\alpha} u^{\frac{1}{\alpha}-1} du = dx$ so that

$$2 \int_0^\infty e^{-x^\alpha} dx = 2 \int_0^\infty \frac{1}{\alpha} u^{\frac{1}{\alpha}-1} e^{-u} du = \frac{2}{\alpha} \Gamma\left(\frac{1}{\alpha}\right).$$

This proves the first claim. To show the second claim, recall that the Fisher Information is also equal to

$$I_f = \mathbb{E}\left[\left(\frac{f'(s)}{f(s)}\right)^2\right] = -\mathbb{E}\left[\frac{\partial^2}{\partial s^2} \log(f(s))\right].$$

This means that it is the case that

$$\begin{aligned}
I_{f_\alpha} =& \frac{\alpha}{2\Gamma(1/\alpha)} \int_{-\infty}^{\infty} \left(\frac{\partial^2}{\partial x^2} |x|^\alpha \right) e^{-|x|^\alpha} dx = \frac{\alpha}{\Gamma(1/\alpha)} \int_0^\infty \left(\frac{\partial^2}{\partial x^2} x^\alpha \right) e^{-x^\alpha} dx \\
=& \frac{\alpha}{\Gamma(1/\alpha)} \int_0^\infty \alpha(\alpha-1) x^{\alpha-2} e^{-x^\alpha} dx.
\end{aligned}$$

Now, perform again the substitution of $u = x^\alpha$.

$$\begin{aligned}
=& \frac{\alpha}{\Gamma(1/\alpha)} \int_0^\infty \alpha(\alpha-1) \frac{1}{\alpha} u \cdot u^{-\frac{2}{\alpha}} \cdot u^{\frac{1}{\alpha}-1} e^{-u} dx \\
=& \frac{\alpha}{\Gamma(1/\alpha)} \int_0^\infty (\alpha-1) u^{-\frac{1}{\alpha}} e^{-u} dx = \frac{\alpha(\alpha-1)}{\Gamma(1/\alpha)} \Gamma\left(1 - \frac{1}{\alpha}\right).
\end{aligned}$$

Since it is the case that $\Gamma(\alpha+1) = \alpha\Gamma(\alpha)$,

$$\Gamma\left(1 - \frac{1}{\alpha}\right) = \frac{\alpha}{\alpha - 1} \Gamma\left(2 - \frac{1}{\alpha}\right),$$

Concluding,

$$I_{f_\alpha}(0) = \frac{\alpha^2}{\Gamma(1/\alpha)} \Gamma\left(2 - \frac{1}{\alpha}\right).$$

$\square$

In all proofs, denote $\kappa_\alpha(\tau)$ by $\kappa(\alpha;\tau)$. This is just done for clarity: κ is a function of α.

Proof of Lemma 2. First, a simple computation shows that $\kappa(\alpha; 0) = \Gamma\left(\frac{1}{\alpha}\right)\Gamma\left(2 - \frac{1}{\alpha}\right)$. This shows that $\kappa(1; 0) = 1$, as $\Gamma(1) = 1$.

I now proceed to the other claims. Let $\psi(\alpha)$ denote the digamma function

$$\psi(\alpha) := \frac{\Gamma'(\alpha)}{\Gamma(\alpha)}.$$

It is well-known that ψ is strictly increasing on $(0, \infty)$.

Now, let $\kappa'(\alpha;0)$ denote $\frac{\partial}{\partial\alpha}\kappa(\alpha;0)$. A straightforward computation yields that

$$\kappa'(\alpha;0)=\frac{\Gamma\left(\frac{1}{\alpha}\right)\Gamma\left(2-\frac{1}{\alpha}\right)}{\alpha^2}\left(\psi\left(2-\frac{1}{\alpha}\right)-\psi\left(\frac{1}{\alpha}\right)\right).$$

Since ψ is strictly increasing, it is immediately seen that $\kappa'(\alpha;0)>0$.

It is left to show that $\lim_{\alpha\to\infty}\kappa(\alpha;0)=\infty$. First, it can be easily verified that $\psi(2)<1$ and $\lim_{\alpha\to\infty}\frac{1}{\alpha}\psi\left(\frac{1}{\alpha}\right)=-1$. These two facts together show that κ' can be seen as

$$\frac{\kappa'(\alpha;0)}{\kappa(\alpha;0)}\approx\frac{1}{\alpha}.$$

That is, $\lim_{\alpha\to\infty}\alpha\frac{\kappa'(\alpha;0)}{\kappa(\alpha;0)}=1$. This means that $\frac{\kappa(\alpha;0)}{\alpha}\to 1$, and hence $\kappa'(\alpha;0)\to 1$. This shows that κ diverges, completing the proof. □

Proof of Proposition 1. As noted, I restrict attention to $\tau\in[0,\infty)$ for the proof. The exact analogues apply for $\tau\leq 0$.

For ease of reference I restate that the objective that the platform aims to maximize is given by.

$$g_\alpha(\tau):=\frac{f_\alpha(\tau)^2}{F_\alpha(\tau)(1-F_\alpha(\tau))} \tag{7}$$

I proceed in several steps. I first show that $\tau^*(\alpha)\neq 0$ for $\alpha>2$, and that it 0 is a candidate for $\tau^*(\alpha)$ for $\alpha\leq 2$.

Lemma 5. *It is the case that* 0 *is a local maximum of* $g_\alpha(\tau)$ *for* $\alpha\leq 2$*, and a local minimum of* $g_\alpha(\tau)$ *for* $\alpha>2$*.*

Proof of Lemma 5. The derivative of (7) with respect to τ is given by (dropping the subscript α)

$$\frac{f(\tau)}{F(\tau)(1-F(\tau))}\left[2f'(\tau)-f(\tau)^2\left(\frac{1}{F(\tau)}-\frac{1}{1-F(\tau)}\right)\right].$$

As f is of full support, $\frac{f(\tau)}{F(\tau)(1-F(\tau))}>0$, and so can be ignored when searching for a critical value, and hence a maximum. The first order condition to be a local maximum then becomes

$$2f'(\tau) = f(\tau)^2 \left(\frac{1}{F(\tau)} - \frac{1}{1 - F(\tau)} \right). \tag{8}$$

For any $\alpha > 1$, 0 will be a solution to (8), and hence a candidate for a maximizer. This is as $f'_\alpha(0) = 0$ and $F_\alpha(0) = \frac{1}{2}$.

This means that to verify the claim the second order condition is needed. At 0, it can be verified that the second-order condition for a maximizer (since $F_\alpha(0) = \frac{1}{2}$) is

$$\frac{1}{2} f_\alpha(0)^4 + \frac{1}{8} f'_\alpha(0)^2 + \frac{1}{8} f_\alpha(0) f''_\alpha(0) \leq 0.$$

Notice, for all $\alpha > 2$, we have that $f'_\alpha(0) = 0$ and $f''_\alpha(0) = 0$. This means that, as $f_\alpha(0) > 0$, for all $\alpha > 2$, $\tau = 0$ is a local minimum, and hence is not optimal. Similarly, for $\alpha < 2$, it is the case that $\lim_{\tau \to 0} f''_\alpha(\tau) = -\infty$, while $f'_\alpha(0), f_\alpha(0) > 0$ This means that 0 is indeed a local maximum for $\alpha < 2$. The case $\alpha = 2$ can be computed numerically, and it can be seen that this is indeed a local maximum in this case. $\square$

I now show that there is at most one possible candidate for the maximizer of $g_\alpha(\tau)$ on $\tau \in [0, \infty)$. This will show uniqueness of $\tau^*(\alpha)$.

Lemma 6. *The following are true:*

(i) For $\alpha > 2$, there is exactly one value of $\tau \in (0, \infty)$ that satisfies (8) and it is a local maximum of $g_\alpha(\tau)$; and

(ii) For $\alpha \leq 2$, no value of $\tau \in (0, \infty)$ satisfies (8).

Proof of Lemma 6. Throughout I drop the subscript of α for notational clarity.

Case 1: $\alpha > 2$.

Straightforward algebra allows (8) to be rewritten as

$$\frac{2f'(\tau)}{1 - 2F(\tau)} = \frac{f^2(\tau)}{F(\tau)(1 - F(\tau))}. \tag{9}$$

Notice that the right-hand side of (9) is exactly $g(\tau)$. This means that the value of g can also be used to determine the sign of its derivative. Specifically, the critical points of the objective are exactly those that intersect the curve given by

$$w(\tau) := \frac{2f'(\tau)}{1-2F(\tau)}.$$

It is exactly when $g(\tau) = w(\tau)$ that $g'(\tau) = 0$. Moreover,

$$g'(\tau) > 0 \iff g(\tau) > w(\tau).$$

Due to this relationship, the characteristics of $w(\tau)$ imply certain characteristics of $g(\tau)$. In particular, I now show that $w(\tau)$ has one unique maximum on $(0, \infty)$. To do this, I iterate on the process above. The sign of $w'(\tau) = 0$ is determined by the sign of

$$2f''(\tau)(1-2F(\tau)) + 4f'(\tau)f(\tau) \tag{10}$$

in particular, $w'(\tau) = 0$ if and only if

$$\frac{2f'(\tau)}{1-2F(\tau)} = -\frac{f''(\tau)}{f(\tau)} =: h(\tau).$$

Moreover, $w'(\tau) > 0$ if and only if $h(\tau) > w(\tau)$ (notice that this is the reverse relationship of w and g).

While w and g are difficult to manage because the presence of F in their definition (and there is no closed-form expression for the incomplete Gamma functions), $h(\tau) = (\alpha^2 - \alpha)x^{\alpha-2} - \alpha^2 x^{2\alpha-2}$. This (relative to g and w) is an extremely simple function. It can be seen that for all $\alpha > 2$, h has the following properties: (i) $h(0) = 0$, and $h(\tau) > 0$ for τ in a neighbourhood of 0; (ii) h is single-peaked; (iii) $h(1) < 0$; and (iv) $h(\tau) > w(\tau)$ in a neighbourhood of 0. Properties (i)-(iii) are easily to verify, and property (iv) follows from $w'(\tau) > 0$ in a neighbourhood of 0 (which directly implies that $w(\tau) < h(\tau)$).

Claim 1a: $w(\tau)$ has at one critical value on $(0, \infty)$. It is a local maximum, and this critical value is on $(0, 1)$.

Recall that if $w(\tau) < h(\tau)$, then $w'(\tau) > 0$, and if $w(\tau) > h(\tau)$, then $w'(\tau) < 0$. First, observe that all critical values of w must be on $(0, 1)$ because h is negative of

$[1, \infty)$ (so $w(\tau) \neq h(\tau)$ for any $\tau \geq 1$).

Second, let τ_h be the value of τ that maximizes h. Notice that h is increasing on $(0, \tau_h)$ and decreasing on (τ_h, ∞). Suppose now that $w(\tau') = h(\tau')$ for some $\tau' \in (0, \tau_h)$. There must be smallest value of τ' for which this holds, as $w(\tau)$ is strictly increasing in a neighbourhood of 0. Then for this τ', it is the case that $w(\tau') = 0$ and $h(\tau') > 0$. This means that for some $\epsilon > 0$ it is the case that for all $\tau \in (\tau' - \epsilon, \tau')$ it is the case that $w(\tau) > h(\tau)$. But τ' is the smallest positive value where the two cross, which is a contradiction. This means that it cannot be the case that $w'(\tau) = 0$ when $h'(\tau) > 0$. This is as, for w to cross h when h is increasing it must be the case that w is larger than h before the crossing.

Since eventually $w(\tau) > h(\tau)$, it must be the case then that either $w = h$ at the peak of h, or on the region where h is decreasing. In either case it must be the case that w goes from increasing to decreasing, and can not cross h again. This proves the claim.

Claim 1b: $g(\tau)$ has a most one critical value on $(0, \infty)$. It is a local maximum, and this critical value is on $(0, 1)$.

Recall that, if $g(\tau) > w(\tau)$, then $g'(\tau) > 0$, and if $g(\tau) < w(\tau)$, then $g'(\tau) < 0$. Notice that because zero is a local minimum of $g(\tau)$ (by Lemma 5), $g(\tau)$ is increasing in a neighbourhood of zero (since we are concerned only with $\tau \geq 0$. Moreover, it is the case that $g(0) > 0 = w(0)$. This second equality holds because $f''(0) = 0$ for $\alpha > 2$. Denote by τ_w the value of τ that maximizes w on $(0, \infty)$.

Now, suppose that $g(\tau') = w(\tau')$ for some $\tau' \in (\tau_w, \infty)$. This is because it implies that $g(\tau) > w(\tau)$ for $\tau = (\tau', \tau' + \epsilon)$ for some $\epsilon > 0$. However, this means that $g'(\tau) > 0$ on this region. Because w is decreasing on this region, it implies that $g'(\tau) > 0$ on (τ', ∞), contradicting that $\lim_{\tau \to \infty} g(\tau) = 0$. This means that the two cannot intersect when $w'(\tau) < 0$.

By a similar logic, it must be the case that $g(\tau) = w(\tau)$ for some $\tau \in (0, \tau_w]$. Call the first point where they intersect τ^* (well-defined because they are not equal at 0). Notice that for all $\tau \in (\tau^*, \tau^* + \epsilon)$, it must be the case that g is decreasing.[22] But, g must continue to decrease until they intersect again. So, they cannot intersect until

[22] This is obvious in the case that $\tau^* < \tau_w$. In the case that $\tau^* = \tau_w$, it follows from a similar argument to showing that there is no intersection when w is decreasing.

w decreases. We have, however, ruled out intersection in that case, so it must be that τ^* is unique, proving the claim and finishing Case 1.

Case 2: $\alpha < 2$.
In the case of $\alpha < 2$ showing that there is no critical value on $(0, \infty)$ is simpler and so I omit the details.

In this case, (i) $\lim_{\tau \to 0} w(\tau) = \infty$ and (ii) $h(\tau)$ is monotonically decreasing. These jointly imply that $w(\tau)$ is monotonically decreasing to 0. This in turn implies that $g(\tau)$ is monotonically decreasing to zero (since they cannot intersect when w decreases in a similar argument to above). This shows that in the case that $\alpha < 2$ it must be that $\tau = 0$ is the only possible critical value.

Case 3: $\alpha = 2$.
This is dealt with in a similar manner to Case 2. The major difference is that $w(\tau)$ does not diverge at 0. Instead, it is easy to show that $w(0) = h(0)$ and w is decreasing in a region of 0, so that it must be monotonically decreasing. This case then reduces to Case 2. $\square$

Now that I have showed that τ^* exists and is unique, it is sufficient to look at the solution to the first order condition (8) on $(0, \infty)$. The last things to check are that $\tau^*(\alpha) \to 1$, and the bounding value of κ.

Claim: $\lim_{\alpha \to \infty} \tau^*(\alpha) = 1$.

First, notice that $\tau^*(\alpha) < 1$ for all $\alpha \geq 1$, by Lemma 6.

So, I simply show that for all $\tau < 1$, there exists an α_τ such that: $\alpha \geq \alpha_\tau$, $g'_\alpha(\tau) > 0$. To do this, explicitly write the inner term

$$\begin{aligned}
&2f'(\tau) - f(\tau)^2 \left(\frac{1}{F(\tau)} - \frac{1}{1 - F(\tau)} \right) \\
&\qquad = -\alpha \tau^{\alpha - 1} \cdot \frac{\alpha}{\Gamma(1/\alpha)} e^{-\tau^\alpha} \\
&\qquad\quad + \left(\frac{\alpha}{2\Gamma(1/\alpha)} e^{-\tau^\alpha} \right)^2 \left[\frac{\Gamma(1/\alpha)}{\frac{1}{2}\Gamma(1/\alpha, \tau^\alpha)} - \frac{\Gamma(1/\alpha)}{\frac{1}{2}\Gamma(1/\alpha) + \frac{1}{2}\gamma(1/\alpha, \tau^\alpha)} \right],
\end{aligned}$$

where here $\Gamma(1/\alpha, \cdot)$ and $\gamma(1/\alpha, \cdot)$ are the upper- and lower-incomplete gamma functions, respectively. Because we are only interested in the sign of this expression, it is possible to simplify. It is equivalent to look at the sign of

$$-2\tau^{\alpha-1} + e^{-\tau^\alpha}\left[\frac{1}{\Gamma(1/\alpha,\tau^\alpha)} - \frac{1}{\Gamma(1/\alpha)+\gamma(1/\alpha,\tau^\alpha)}\right]. \tag{11}$$

That is, $g'_\alpha(\tau) > 0$ exactly when (11) is positive. When $\tau < 1$, the first term converges to 0, so we need only ensure that the second term stays bounded from 0. First, for fixed $\tau < 1$, $e^{-\tau^\alpha} \to 1$. Second, as $\alpha \to \infty$, it is easy to verify that $f_\alpha(\tau) \to \frac{1}{2}\mathbb{1}\{\tau \in [-1,1]\}$ (except for the measure zero set of $\{-1,1\}$). This means that

$$\lim_{\alpha\to\infty}\left(\frac{1}{F(\tau)} - \frac{1}{1-F(\tau)}\right) = \left(\frac{2}{\tau+1} - \frac{2}{1-\tau}\right) > 0$$

for all fixed $\tau \in (0,1)$. This means that neither portions of the second term go to zero, so that for any τ in this interval, $g'_\alpha(\tau)$ is eventually positive.

This means that $\tau^*(\alpha)$ must eventually grow to 1. It is not hard to use the same arguments to show that $\tau^*(\alpha)^\alpha$ converges to the solution of $\Gamma(0,x) - \frac{1}{2x}e^{-x} = 0$.

Claim: $\kappa(\alpha;\tau^*(\alpha)) < 2e^2\int_1^\infty \frac{e^{-x}}{x}dx$.

For all $\alpha \geq 4$, it can be seen that $\kappa(\alpha,1)$ is less than this limit. For $\alpha < 4$, using the naive threshold of 0 will also return a value that is less than this limit. This shows the final claim, and completes the proof of Proposition 1. □

Proof of Proposition 2. In order to prove Proposition 2(i), I prove the following, stronger, lemma.

Lemma 7. *$\tau^*(\alpha)^\alpha$ is strictly increasing in α for $\alpha \geq 2$.*

Proof of Lemma 7. First, notice that

$$\frac{\partial\tau^*(\alpha)^\alpha}{\partial\alpha} > 0 \iff \left(\frac{\partial^2}{\partial\tau^\alpha\partial\alpha}\frac{f_\alpha^2(\tau)}{F_\alpha(\tau)(1-F_\alpha(\tau))}\right)\Big|_{\tau=\tau^*(\alpha)} > 0.$$

That is, letting $x = \tau^\alpha$, the sign of

$$\frac{\partial}{\partial\alpha}\left[-2\frac{x}{x^{1/\alpha}}e^x + \left[\frac{1}{\Gamma(1/\alpha,x)} - \frac{1}{\Gamma(1/\alpha)+\gamma(1/\alpha,x)}\right]\right] \tag{12}$$

must be determined at the optimal value of $x^*(\alpha) = \tau^*(\alpha)^\alpha$. Notice that (12) is not exactly equal to the derivative, because there is an additional positive term that multiplies this. However, because what is in the brackets here is zero at the optimal threshold, this term can be ignored (because it will not influence sign of this object). This just follows from the envelope theorem. So, we must show that (12) is positive at $x^*(\alpha)$.

Another simplification is to replace α with $1/\alpha$. That is, let $\beta := 1/\alpha$. A straightforward computation that yields that the sign of (12) is the opposite of the sign of

$$\begin{aligned} b_\beta(x) :=& 2\log(x)\frac{x}{x^\beta}e^x \\ & - \frac{\int_x^\infty y^{\beta-1}\log(y)e^{-y}dy}{\Gamma(\beta,x)^2} + \frac{\int_0^\infty y^{\beta-1}\log(y)e^{-y}dy + \int_0^x y^{\beta-1}\log(y)e^{-y}dy}{(\Gamma(\beta)+\gamma(\beta,x))^2}. \end{aligned}$$

This is because $\frac{\partial}{\partial\alpha}\beta < 0$. So, I show that $b_\beta(x^*(\beta)) < 0$ (abusing notation here writing $x^*(\beta)$) for $\beta < \frac{1}{2}$. As a final choice to make notation easier, let λ_β denote the measure that is induced by $y^{\beta-1}e^{-y}$ (that is, $d\lambda_\beta(y) = y^{\beta-1}e^{-y}dy$). Then, $b_\beta(x)$ can be written as

$$2\log(x)\frac{x}{x^\beta}e^x - \frac{\int_x^\infty \log(y)d\lambda_\beta(y)}{\Gamma(\beta,x)^2} + \frac{\int_0^\infty \log(y)d\lambda_\beta(y) + \int_0^x \log(y)d\lambda_\beta(y)}{(\Gamma(\beta)+\gamma(\beta,x))^2}. \tag{13}$$

Note that in the notation above, we have that

$$\Gamma(\beta) = \int_0^\infty d\lambda_\beta(y), \quad \Gamma(\beta,x) = \int_x^\infty d\lambda_\beta(y), \quad \text{and} \quad \gamma(\beta) = \int_0^x d\lambda_\beta(y).$$

Now, at $x^*(\beta)$, from (12), it is the case that

$$2x^{1-\beta}e^x = \frac{1}{\Gamma(\beta,x)} - \frac{1}{\Gamma(\beta)+\gamma(\beta,x)}.$$

This means that it is possible to write (13) at $x^*(\beta)$ (I write x instead of $x^*(\beta)$ for conciseness) as

$$\begin{aligned} &\log(x)\left(\frac{1}{\Gamma(\beta,x)} - \frac{1}{\Gamma(\beta)+\gamma(\beta,x)}\right) \\ &\quad - \frac{\int_x^\infty \log(y)d\lambda_\beta(y)}{\Gamma(\beta,x)^2} + \frac{\int_0^\infty \log(y)d\lambda_\beta(y) + \int_0^x \log(y)d\lambda_\beta(y)}{(\Gamma(\beta)+\gamma(\beta,x))^2}. \end{aligned}$$

$$
\begin{aligned}
=&\frac{\log(x)\int_x^\infty d\lambda_\beta(y)}{\Gamma(\beta,x)^2}-\frac{\log(x)\left(\int_0^\infty d\lambda_\beta(y)+\int_0^x d\lambda_\beta(y)\right)}{\Gamma(\beta)+\gamma(\beta,x)}\\
&-\frac{\int_x^\infty \log(y)d\lambda_\beta(y)}{\Gamma(\beta,x)^2}+\frac{\int_0^\infty \log(y)d\lambda_\beta(y)+\int_0^x \log(y)d\lambda_\beta(y)}{(\Gamma(\beta)+\gamma(\beta,x))^2}\\
=&\frac{\int_x^\infty \left(\log(x)-\log(y)\right)d\lambda_\beta(y)}{\Gamma(\beta,x)^2}\\
&+\frac{\int_0^\infty \left(\log(y)-\log(x)\right)d\lambda_\beta(y)+\int_0^x \left(\log(y)-\log(x)\right)d\lambda_\beta(y)}{(\Gamma(\beta)+\gamma(\beta,x))^2}.
\end{aligned}
$$

Since log is an increasing function, it must be that $\int_x^\infty \left(\log(x)-\log(y)\right)d\lambda_\beta(y)<0$ whenever $\beta<\frac{1}{2}$, so that $x^*(\beta)>0$. Moreover, it must always be the case that $\Gamma(\beta,x)<\Gamma(\beta)+\gamma(\beta,x)$ whenever $x>0$. This means that at $x^*(\beta)$ it is the case that

$$
\frac{\int_x^\infty \left(\log(x)-\log(y)\right)d\lambda_\beta(y)}{\Gamma(\beta,x)^2}<\frac{\int_x^\infty \left(\log(x)-\log(y)\right)d\lambda_\beta(y)}{\left(\Gamma(\beta)+\gamma(\beta,x)\right)^2}.
$$

This means that (13) evaluated at $x^*(\beta)$ is less than

$$
\begin{aligned}
&\frac{\int_x^\infty \left(\log(x)-\log(y)\right)d\lambda_\beta(y)}{(\Gamma(\beta)+\gamma(\beta,x))^2}\\
&+\frac{\int_0^\infty \left(\log(y)-\log(x)\right)d\lambda_\beta(y)+\int_0^x \left(\log(y)-\log(x)\right)d\lambda_\beta(y)}{(\Gamma(\beta)+\gamma(\beta,x))^2}\\
=&\frac{2\int_0^x \left(\log(y)-\log(x)\right)d\lambda_\beta(y)}{(\Gamma(\beta)+\gamma(\beta,x))^2}<0.
\end{aligned}
$$

This proves that $x^*(\beta)$ is decreasing in β, so that $\tau^*(\alpha)^\alpha$ is increasing in α for all $\beta<\frac{1}{2}$. This in turn shows that $\tau^*(\alpha)^\alpha$ is increasing in α for all $\alpha\geq 2$. □

As noted, Proposition 2(i) follows as an immediate corollary of this lemma, since $\tau^*(\alpha)<1$ for all α.

Proof of Proposition 2(ii):

In what follows, let $P(\alpha,t):=\gamma(\alpha,t)/\Gamma(\alpha)$ denote the *normalized lower incomplete gamma function.*

First, recall that $\tau^*(\alpha)^\alpha$ strictly increasing in α by Lemma 7.

I show the claim separately on two intervals, based off of the behaviour of $f_\alpha(0)$ as a function of α. In what follows, let α^* be defined as the unique value for which $\psi(1/\alpha) + \alpha = 0$. For $\alpha < \alpha^*$, it is the case that $\Gamma(1/\alpha)/\alpha$ is decreasing in α (so that $f_\alpha(0)$ is increasing), and for $\alpha > \alpha^*$ it is the case the $\Gamma(1/\alpha)/\alpha$ is increasing in α (so that $f_\alpha(0)$ is decreasing).

Case 1: $\alpha < \alpha^*$.
First, on this region, for a fixed τ with $0 < \tau < 1$, it must be the case that $f_\alpha(\tau)$ is increasing in α. This is because clearly τ^α is decreasing in α, and this is the region where $\frac{\alpha}{\Gamma(\alpha)}$ is increasing.

Now, $F_\alpha(0) \equiv \frac{1}{2}$, and for all $0 < \tau < \tau'$, it is the case that $f_\alpha(\tau)$ is increasing in α. This means that, for all $\alpha, \alpha' \in (2, \alpha^*)$ with $\alpha < \alpha'$, it is the case that

$$F_\alpha(\tau^*(\alpha)) < F_{\alpha'}(\tau^*(\alpha))$$

From Lemma 7 it is the case that $\tau^*(\alpha') > \tau^*(\alpha)$, so that

$$F_{\alpha'}(\tau^*(\alpha)) < F_{\alpha'}(\tau^*(\alpha'))$$

This shows that $F_\alpha(\tau^*(\alpha))$ is increasing in α on this range, completing the proof in this case.

Case 2: $\alpha > \alpha^*$.
Since we have that $\alpha > \alpha^*$ and we have showed that $\tau^*(\alpha)$ is increasing in α, it must be the case that $f_\alpha(\tau^*(\alpha))$ is decreasing in α for $\alpha > \alpha^*$, as by definition $\Gamma(1/\alpha)/\alpha$ is increasing for all $\alpha > \alpha^*$.

This means that if it is the case that

$$\frac{f_\alpha(\tau^*(\alpha))^2}{F_\alpha(\tau^*(\alpha))(1 - F_\alpha(\tau^*(\alpha)))} \tag{14}$$

is increasing in α, then the claim will hold in this case. This is because, for (14) to be increasing, if the numerator is decreasing then the denominator must be in turn decreasing. This implies that $F_\alpha(\tau^*(\alpha))$ is increasing in α, which is what we set out to show. A stronger condition, which is what we will prove, is that

$$\frac{\left(\frac{\alpha}{\Gamma(1/\alpha)}\right)^2}{(1+P(1/\alpha,\tau))(1-P(1/\alpha,\tau))} \tag{15}$$

is increasing in α for fixed τ, where $P(1/\alpha,\tau) := \frac{\Gamma(1/\alpha,\tau)}{\Gamma(1/\alpha)}$ is the normalized incomplete Gamma function. This is a stronger condition because it is saying is that if we look at values of τ for each α that make $e^{-\tau^\alpha}$ equal, then these values of τ cause this ratio to increase. In particular, consider evaluating (14) at $\tau^*(\alpha)$ for some α. If (15) is increasing in α, then for all $\alpha' > \alpha$ it must be the case that

$$\frac{f_{\alpha'}(\tau^*(\alpha)^{\alpha/\alpha'})^2}{F_{\alpha'}(\tau^*(\alpha)^{\alpha/\alpha'})(1-F_{\alpha'}(\tau^*(\alpha)^{\alpha/\alpha'}))} > \frac{f_\alpha(\tau^*(\alpha))^2}{F_\alpha(\tau^*(\alpha))(1-F_\alpha(\tau^*(\alpha)))}.$$

Since by optimality it is the case that

$$\frac{f_{\alpha'}(\tau^*(\alpha'))^2}{F_{\alpha'}(\tau^*(\alpha'))(1-F_{\alpha'}(\tau^*(\alpha')))} \geq \frac{f_{\alpha'}(\tau^*(\alpha)^{\alpha/\alpha'})^2}{F_{\alpha'}(\tau^*(\alpha)^{\alpha/\alpha'})(1-F_{\alpha'}(\tau^*(\alpha)^{\alpha/\alpha'}))},$$

this will prove the claim.

Now, to show that (15) is indeed increasing in α. Perform the switch to $\beta := 1/\alpha$, as the in proof of Lemma 7. Then, the above can be rewritten as

$$\begin{aligned}&\frac{\left(\frac{1}{\beta\Gamma(\beta)}\right)^2}{(1+P(\beta,x))(1-P(\beta,x))}\\ &= \frac{1}{2\beta^2\Gamma(\beta)}\left(\frac{1}{\Gamma(\beta)+\gamma(\beta,x)}+\frac{1}{\Gamma(\beta)-\gamma(\beta,x)}\right).\end{aligned}$$

Multiplying by 2 and then taking the derivative with respect to β here yields

$$\begin{aligned}&-\frac{2\beta\Gamma(\beta)+\beta^2\int_0^\infty \log(y)d\lambda_\beta(y)}{(\beta^2\Gamma(\beta))^2}\left(\frac{1}{\Gamma(\beta)+\gamma(\beta,x)}+\frac{1}{\Gamma(\beta)-\gamma(\beta,x)}\right)\\ &-\frac{1}{\beta^2\Gamma(\beta)}\left(\frac{\int_0^\infty \log(y)d\lambda_\beta(y)+\int_0^x \log(y)d\lambda_\beta(y)}{(\Gamma(\beta)+\gamma(\beta,x))^2}+\frac{\int_x^\infty \log(y)d\lambda_\beta(y)}{(\Gamma(\beta)-\gamma(\beta,x))^2}\right).\end{aligned}$$

Here I use the definition of $\lambda_\beta(y)$ from the proof of Lemma 7. The goal is to show that this object is negative. Again let $\Gamma'(\beta)/\Gamma(\beta) =: \psi(\beta)$ and observe that for $\alpha > \alpha^*$ by definition it the case that $\frac{1}{\beta} + \psi(\beta) < 0$. Now, straightforward arithmetic shows that the sign of this is equivalent to the sign of

$$-\frac{\int_0^\infty \left(\log(y) + \frac{2}{\beta} + \psi(\beta)\right) d\lambda_\beta(y) + \int_0^x \left(\log(y) + \frac{2}{\beta} + \psi(\beta)\right) d\lambda_\beta(y)}{(\Gamma(\beta) + \gamma(\beta, x))^2}$$
$$- \frac{\int_x^\infty \left(\log(y) + \frac{2}{\beta} + \psi(\beta)\right) d\lambda_\beta(y)}{(\Gamma(\beta) - \gamma(\beta, x))^2}.$$

I first show that for all $x \in (0, \infty)$ and β, it is the case that.

$$-\int_x^\infty \left(\log(y) + \frac{2}{\beta} + \psi(\beta)\right) d\lambda_\beta(y) < 0.$$

Now, consider that on this region of β, by definition it is the case that

$$-\left(\frac{2}{\beta} + \psi(\beta)\right) = 2\left(\frac{1}{\beta} + \psi(\beta)\right) + \psi(\beta) < \psi(\beta).$$

This means that

$$-\int_x^\infty \left(\log(y) + \frac{2}{\beta} + \psi(\beta)\right) d\lambda_\beta(y) < \int_x^\infty \left(-\log(y) + \psi(\beta)\right) d\lambda_\beta(y)$$

Now, the sign of the above is equivalent to the sign of

$$\frac{\int_x^\infty \left(-\log(y) + \psi(\beta)\right) d\lambda_\beta(y)}{\int_x^\infty d\lambda_\beta(y)}.$$

This, however, is clearly decreasing in x, because smaller values of x correspond to the average log value being smaller. Since it is clearly equal to zero at $x = 0$, this shows that

$$-\int_x^\infty \left(\log(y) + \frac{2}{\beta} + \psi(\beta)\right) d\lambda_\beta(y) < 0.$$

This in turn implies that

$$-\frac{\int_x^\infty \left(\log(y) + \frac{2}{\beta} + \psi(\beta)\right) d\lambda_\beta(y)}{(\Gamma(\beta) - \gamma(\beta, x))^2} < -\frac{\int_x^\infty \left(\log(y) + \frac{2}{\beta} + \psi(\beta)\right) d\lambda_\beta(y)}{(\Gamma(\beta) + \gamma(\beta, x))^2}.$$

This means that it is sufficient to check the sign of

$$
\begin{aligned}
&-\int_0^\infty \left(\log(y)+\frac{2}{\beta}+\psi(\beta)\right) d\lambda_\beta(y) - \int_0^x \left(\log(y)+\frac{2}{\beta}+\psi(\beta)\right) d\lambda_\beta(y) \\
&\quad - \int_x^\infty \left(\log(y)+\frac{2}{\beta}+\psi(\beta)\right) d\lambda_\beta(y)
\end{aligned}
$$

and ensure that it is negative. But this expression is equal to

$$
-2\int_0^\infty \left(\log(y)+\frac{2}{\beta}+\psi(\beta)\right) d\lambda_\beta(y) = -\Gamma'(\beta) - \left(\frac{2}{\beta}+\psi(\beta)\right)\Gamma(\beta) < 0,
$$

where this last inequality follows from the observation above:

$$
\left(\frac{2}{\beta}+\psi(\beta)\right)\Gamma(\beta) > -\Gamma'(\beta)
$$

This completes the proof in this case. $\square$

Proof of Proposition 3. First, notice that

$$
f(s) := f_{\alpha_L,\alpha_H}(s) = \frac{1}{\frac{\Gamma(1/\alpha_L)}{\alpha_L} + \frac{\Gamma(1/\alpha_H)}{\alpha_H}} \left(\mathbb{1}\{s \geq 0\}e^{-s^{\alpha_H}} + \mathbb{1}\{s \leq 0\}e^{-(-s)^{\alpha_L}}\right)
$$

Recall that, by definition, $\frac{\Gamma(1/\alpha)}{\alpha}$ is increasing in α for all $\alpha \geq \alpha^*$. This means, in particular, that there is a larger mass of positive signals than negative signals in this case.

What must be shown is that

$$
\frac{f(\tau)^2}{F(\tau)(1-F(\tau))}
$$

is maximized for a negative value of τ. For ease of notation, let $\alpha := \alpha_H$ and $\alpha' := \alpha_L$. Notice that as before, the numerator is strictly increasing in τ for $\tau < 0$, and strictly decreasing in τ for $\tau > 0$. Since $\alpha' > \alpha$, for small values of τ there is larger persistence in the value of f on the negative side of the distribution.[23] Explicitly, given a $\tau > 0$, we will have that $-\tau^{\alpha/\alpha'} < -\tau$ (for $\tau < 1$, which will again be optimal here) will

[23]Of course, for $\tau > 1$ this is reversed, because the tails are smaller on the negative side of the distribution than the positive side of the distribution.

have the same value of the numerator. From here, the claim is two-fold.

Claim 1: The optimal choice of τ for $\tau > 0$ has $\tau \geq \tau^*(\alpha)$.

To see this, consider that the first order condition is given by

$$-2\alpha\tau^{\alpha-1} - e^{-x^\alpha}\left[\frac{1}{\frac{\Gamma(1/\alpha')}{\alpha'} + \int_0^\tau e^{-x^\alpha}dx} - \frac{1}{\int_\tau^\infty e^{-x^\alpha}dx}\right].$$

As $\frac{\Gamma(1/\alpha')}{\alpha'} > \frac{\Gamma(1/\alpha)}{\alpha}$, the first fraction is smaller, so that the positive component is larger. This means that for the first order condition to be satisfied for $\tau > 0$ we need $\tau > \tau^*(\alpha)$.

Claim 2: For all $\tau > \tau^*(\alpha)$, it is the case that $1 - F(\tau) > F(-\tau^{\alpha/\alpha'})$.

Notice that

$$1 - F(\tau) = \frac{1}{\frac{\Gamma(1/\alpha)}{\alpha} + \frac{\Gamma(1/\alpha')}{\alpha'}}\int_\tau^\infty e^{-x^\alpha}dx = \frac{1}{\frac{\Gamma(1/\alpha)}{\alpha} + \frac{\Gamma(1/\alpha')}{\alpha'}} \cdot \frac{\Gamma\left(\frac{1}{\alpha}, \tau^\alpha\right)}{\alpha}$$

and

$$F(-\tau^{\alpha/\alpha'}) = \frac{1}{\frac{\Gamma(1/\alpha)}{\alpha} + \frac{\Gamma(1/\alpha')}{\alpha'}}\int_{\tau^{\alpha/\alpha'}}^\infty e^{-x^{\alpha'}}dx = \frac{1}{\frac{\Gamma(1/\alpha)}{\alpha} + \frac{\Gamma(1/\alpha')}{\alpha'}} \cdot \frac{\Gamma\left(\frac{1}{\alpha'}, \tau^\alpha\right)}{\alpha'}$$

From here it is sufficient to show that $\frac{\Gamma(1/\alpha, \tau^*(\alpha)^\alpha)}{\alpha} > \frac{\Gamma(1/\alpha', \tau^*(\alpha)^\alpha)}{\alpha'}$ for all $\alpha' > \alpha$. Notice that the value of τ at which this is evaluated is important: it does not hold for all τ (indeed, it clearly will not hold for $\tau = 0$).

Subclaim: $\frac{\Gamma(1/\alpha, \tau^\alpha)}{\alpha} > \frac{\Gamma(1/\alpha' \tau^\alpha)}{\alpha'}$ for $\alpha \geq \alpha^*$, $\tau^\alpha \geq \tau^*(\alpha^*)^{\alpha^*} > 0.001$.

Now, do the standard change of variable for $\beta := 1/\alpha$ and $x := \tau^\alpha$. I show that for $\beta < 1/\alpha^*$ it is the case that

$$\frac{\partial}{\partial\beta}\left(\Gamma(\beta, x) \cdot \beta\right) > 0$$

for all $x > 0.001$. The above is equivalent to (after some straightforward algebra in the same vein as above)

$$-\frac{1}{\beta} < \frac{\int_x^\infty \log(y) d\lambda_\beta(y)}{\int_x^\infty d\lambda_\beta(y)} \tag{16}$$

where the $d\lambda_\beta(y)$ notation is the same as in Lemma 7. Notice first that the right-hand side of (16) is (i) increasing in x and (ii) increasing in β. Property (i) holds because is because log is increasing, and so by increasing x the average log value will increase. Property (ii) follows from a similar logic: smaller β put more weight on smaller values of y (and hence smaller values of log).

This means that if (16) holds for $x^* := 0.001$, then the (sub)claim will be proved. From here, several straightforward computations will prove the claim:

$$-2.16 < \frac{\int_{x^*}^\infty \log(y) d\lambda_{1/2.7}(y)}{\int_{x^*}^\infty d\lambda_{1/2.7}(y)}, \tag{17}$$

dealing with $\beta \in \left(\frac{1}{2.7}, \frac{1}{\alpha^*}\right]$. Then,

$$-2.7 < \frac{\int_{x^*}^\infty \log(y) d\lambda_{1/4}(y)}{\int_{x^*}^\infty d\lambda_{1/4}(y)}, \tag{18}$$

dealing with then $\beta \in \left(\frac{1}{4}, \frac{1}{\alpha^*}\right]$. Finally, it is the case that

$$-4 < \frac{\int_{x^*}^\infty \log(y) d\lambda_0(y)}{\int_{x^*}^\infty d\lambda_0(y)}.$$

This means that for all $\beta \in (0, \alpha^*]$ it is the case that (16) holds. This proves Claim 2.

From here, the conclusion is reached by simply putting together Claims 1 and 2: for any possible optimal positive τ, there is a negative threshold that performs strictly better, so the optimal threshold must be negative. $\square$

A.3 Proofs for Section 5.1

Micro-founding Separable Propensities: I suppose now that reviewers are defined by three characteristics. As before, reviewer i receives a conditionally independent signal s_i about the state of the world. Now, however, their willingness to review $w_i = (w_{i,1}, w_{i,2})$ is comprised of two components.

As before, I assume that s_i and w_i are independent, and also that $w_{i,1}$ and $w_{i,2}$ are independent. The reviewer's decision to review now follows a two-stage procedure. First, as before, the reviewer compares $w_{i,1}$ to a *review-dependent* threshold, which captures the heterogeneity in reporting rates across reviews. Let p_r be the probability that a reviewer "passes" this first stage. This proportion p_r can be thought of as the proportion of the population who would ever leave a review of type r: many consumers never leave written reviews, but are open to leaving a five-star review.

Second, the reviewer compares $w_{i,2}$ to a *signal-dependent* threshold. Let $q(s)$ be the probability with which a reviewer with signal s passes this second stage, conditional on passing the first stage, and assume that $q(s)$ is continuous as a function of s. That is, within the population of mass p_r who might leave a review of type r, among those with signal s only proportion $q(s)$ will actually leave the review: not everyone who sometimes writes reviews will always leave a review. Together, these factors combine to mean that the probability that a reviewer with signal s facing review r leaves a review is $p_r q(s)$.

Proof of Theorem 3. I proceed with the computations in many steps. At risk of abusing notation, let

$$f(s, \mu, \lambda) := f(s+\mu)^{\lambda} f(s-\mu)^{1-\lambda}$$

For an abuse of notation, let $p(s) := p(r, s) = p_r q(s)$ when the review function in question is clear. let $\rho(\mu; p, \lambda)$ denote the value of $\rho(\mu; p)$ evaluated at a fixed (but not necessarily the optimal) λ.

It can easily be seen that $\rho(\mu; p), \rho(\mu; r, p) \to 0$. Hence, l'Hopital's rule must be applied. First, we look at $\lim_{\mu \to 0} \frac{\partial}{\partial \mu} \rho(\mu; p)$. The regularity assumptions on f mean that, for a fixed λ, we have that this is given by

$$-\frac{\partial}{\partial \mu} \rho(\mu; p, \lambda) =$$
$$\int \left(\lambda \frac{f'(s+\mu)}{f(s+\mu)} - (1-\lambda) \frac{f'(s-\mu)}{f(s-\mu)} \right) f(s, \mu, \lambda) p(s) ds$$

$$
\begin{aligned}
&+\lambda\frac{\int((1-p(s))f(s+\mu)ds)^{\lambda-1}}{\int((1-p(s))f(s-\mu)ds)^{\lambda-1}}\int((1-p(s))f'(s+\mu)ds)\\
&-(1-\lambda)\frac{\int((1-p(s))f(s+\mu)ds)^{\lambda}}{\int((1-p(s))f(s-\mu)ds)^{\lambda}}\int((1-p(s))f'(s-\mu)ds)\,.
\end{aligned}
$$

Notice that because of the envelope theorem, we need not consider here the derivative with respect to λ. Consider now the limit as $\mu \to 0$. First, $f(s,\mu,\lambda) \to f(s,\mu,\lambda)$. Hence, the limit is equal to

$$
-\lim_{\mu\to 0}\frac{\partial}{\partial\mu}\rho(\mu;p,\lambda) = (2\lambda-1)\int p(s)\ell_f(s)f(s)ds + (2\lambda-1)\int(1-p(s))f'(s)ds.
$$

This is equal to zero because $\ell_f(s)f(s) = f'(s)$, and $\int f'(s)ds = 0$ (because the expectation of the linear score function is zero).

Hence, we must consider the second derivative. Here it can be verified that the derivative with respect to λ will indeed be irrelevant, because this value is zero. This means that we need not consider how λ varies with μ, and again may consider evaluating at λ point-wise.

Because the notation gets so tedious, for readability of the null review terms, let $H_+(\mu,n) := \int(1-p(s))f(s+\mu)ds$ (where n refers to "null"), and similarly $H_-(\mu,n)$. In a similar vein, let $H'_+(\mu,n) := \int(1-p(s))f'(s+\mu)ds$, and similarly $H'_-(\mu,n)$ (and $H''_\pm(\mu,n)$). Computing, we get that

$$
\begin{aligned}
&-\frac{\partial^2}{\partial\mu^2}\rho(\mu;p,\lambda) =\\
&\quad\int\lambda(\lambda-1)\left(\left(\frac{f'(s+\mu)}{f(s+\mu)}\right)^2+\left(\frac{f'(s+\mu)f'(s-\mu)}{f(s+\mu)f(s-\mu)}\right)\right)f(s,\mu,\lambda)p(s)ds\\
&\quad-\int\lambda(1-\lambda)\left(\left(\frac{f'(s-\mu)}{f(s-\mu)}\right)^2+\left(\frac{f'(s-\mu)f'(s+\mu)}{f(s-\mu)f(s+\mu)}\right)\right)f(s,\mu,\lambda)p(s)ds\\
&\quad+\int\left(\lambda\frac{f''(s+\mu)}{f(s+\mu)}+(1-\lambda)\frac{f''(s-\mu)}{f(s-\mu)}\right)f(s,\mu,\lambda)p(s)ds\\
&\quad+\lambda(\lambda-1)\frac{H_+(\mu,n)^{\lambda-1}}{H_-(\mu,n)^{\lambda-1}}\left(\frac{H'_+(\mu,n)}{H_+(\mu,n)}+\frac{H'_-(\mu,n)}{H_-(\mu,n)}\right)H'_+(\mu,n)\\
&\quad-(1-\lambda)\lambda\frac{H_+(\mu,n)^{\lambda}}{H_-(\mu,n)^{\lambda}}\left(\frac{H'_+(\mu,n)}{H_+(\mu,n)}+\frac{H'_-(\mu,n)}{H_-(\mu,n)}\right)H'_-(\mu,n)
\end{aligned}
$$

$$+\lambda\frac{H_+(\mu,n)^{\lambda-1}}{H_-(\mu,n)^{\lambda-1}}H''_+(\mu,n)+(1-\lambda)\frac{H_+(\mu,n)^{\lambda}}{H_-(\mu,n)^{\lambda}}H''_-(\mu,n).$$

We can then see that the limit is given by

$$\begin{aligned}
\lim_{\mu\to 0}\frac{\partial^2}{\partial\mu^2}\rho(\mu;p,\lambda) = & \\
& 4\lambda(1-\lambda)\int\left(\frac{f'(s)}{f(s)}\right)^2 f(s)p(s)+\int f''(s)p(s)ds \\
& +4\lambda(1-\lambda)\frac{H'_+(0,n)^2}{H_+(0,n)}+H''_+(0,n) \\
= & \quad 4\lambda(1-\lambda)\int\left(\frac{f'(s)}{f(s)}\right)^2 f(s)p(s)+\int p(s)f''(s)ds \\
& +4\lambda(1-\lambda)\frac{\left(\int(1-p(s))f'(s)ds\right)^2}{\int(1-p(s))f(s)ds}+\int(1-p(s))f''(s)ds.
\end{aligned}$$

Now, as $\int f''(s)ds = 0$ (because its anti-derivative is f'), $\int f'(s)ds = 0$, and $\int f(s)ds = 1$, this value is equal to the numerator in the statement of the proposition when $\lambda = \frac{1}{2}$. Notice that this value of λ is clearly the maximizer of this object. minimized object is the negative of this, $\lambda = \frac{1}{2}$ will minimize the numerator, so it is the optimal value of λ.

Of course, at this stage it has not been shown that the computations that were computed above correspond to the value of the numerator: the denominator needs to go to 0 at the same rate for this to be satisfied. However, for linearity of the argument it is worthwhile to perform all of the above computations in order. I now turn attention to the denominator. The terms in the denominator are of the following form. From hereon $p(s)$ to refer to $p_r q(s)$. At the end, we shall replace these values with their true values. Define

$$g_{\tilde{r}}(\mu,\lambda) := \left(\int_{\{s:r(s)\in\tilde{r}\}} f(s+\mu)p(s)ds\right)^{\lambda}\left(\int_{\{s:r(s)\in\tilde{r}\}} f(s-\mu)p(s)ds\right)^{1-\lambda}.$$

Similarly, for null reviews we have

$$g_n(\mu,\lambda) := \left(\int f(s+\mu)(1-p(s))ds\right)^{\lambda}\left(\int f(s-\mu)(1-p(s))ds\right)^{1-\lambda}.$$

The value of the denominator for a fixed λ is of the form

$$\rho(\mu; r, p, \lambda) := 1 - \sum_{\tilde{r} \in R_r} g_{\tilde{r}}(\mu, \lambda) - g_n(\mu, \lambda).$$

In order to make computations easier to follow, I compute the limits term by term. Notice that the computations for $g_n(\mu, \lambda)$ have already been computed above. For notational clarity, define

$$H_+(\mu, \tilde{r}) = \int_{\{r(s) \in \tilde{r}\}} f(s + \mu) p(s) ds.$$

and similarly all of the related functions $(H_-(\mu, \tilde{r}), H'_\pm(\mu, \tilde{r}), H''_\pm(\mu, \tilde{r}))$. Notice here that the term is $p(s)$ and not $1 - p(s)$, because these are submitted reviews. Let

$$\frac{\partial}{\partial \mu} g_{\tilde{r}}(\mu, \lambda) = \lambda \frac{H_+(\mu, \tilde{r})^{\lambda - 1}}{H_-(\mu, \tilde{r})^{\lambda - 1}} H'_+(\mu, \tilde{r}) - (1 - \lambda) \frac{H_+(\mu, \tilde{r})^{\lambda}}{H_-(\mu, \tilde{r})^{\lambda}} H'_-(\mu, \tilde{r}).$$

This means that

$$\lim_{\mu \to 0} \frac{\partial}{\partial \mu} g_{\tilde{r}}(\mu, \lambda) = (2\lambda - 1) H'(0, \tilde{r}).$$

Recall that we have already computed the dynamics of $g_n(\mu, \lambda)$ above. This means that, regardless of λ, we have that

$$\begin{aligned} -\lim_{\mu \to 0} \frac{\partial}{\partial \mu} \rho(\mu; r, p, \lambda) =& (2\lambda - 1) \left(\sum_{\tilde{r}} H'_+(0, \tilde{r}) + H'_+(0, n) \right) \\ =& (2\lambda - 1) \left(\sum_{\tilde{r}} \int_{\{r(s) \in \tilde{r}\}} f'(s) p(s) ds + \int f'(s)(1 - p(s)) ds \right) \\ =& (2\lambda - 1) \int f'(s) ds = 0. \end{aligned}$$

The reasons that we need not consider λ are similar to the reasons in the case of the numerator. Because of this, I may focus on point-wise evaluation with respect to λ throughout. In particular, this means that the we must proceed and take another derivative. We have that

$$\begin{aligned}\frac{\partial^2}{\partial\mu^2}g_{\tilde r}(\mu,\lambda) =& \lambda(\lambda-1)\frac{H_+(\mu,\tilde r)^{\lambda-1}}{H_-(\mu,\tilde r)^{\lambda-1}}\left(\frac{H_+'(\mu,\tilde r)}{H_+(\mu,\tilde r)}+\frac{H_-'(\mu,\tilde r)}{H_-(\mu,\tilde r)}\right)H_+'(\mu,\tilde r)\\ &-(1-\lambda)\lambda\frac{H_+(\mu,\tilde r)^{\lambda}}{H_-(\mu,\tilde r)^{\lambda}}\left(\frac{H_+'(\mu,\tilde r)}{H_+(\mu,\tilde r)}+\frac{H_-'(\mu,\tilde r)}{H_-(\mu,\tilde r)}\right)H_-'(\mu,\tilde r)\\ &+\lambda\frac{H_+(\mu,\tilde r)^{\lambda-1}}{H_-(\mu,\tilde r)^{\lambda-1}}H_+''(\mu,\tilde r)+(1-\lambda)\frac{H_+(\mu,\tilde r)^{\lambda}}{H_-(\mu,\tilde r)^{\lambda}}H_-''(\mu,\tilde r).\end{aligned}$$

Notice that these computations are similar to those used for $g_n(\mu,\lambda)$ in the computation of the numerator's second derivative. Note again that we need not consider the variation in λ here.

This means that

$$\lim_{\mu\to 0}\frac{\partial^2}{\partial\mu^2}g_{\tilde r}(\mu,\lambda) = 4\lambda(\lambda-1)\frac{H_+'(0,\tilde r)^2}{H_+(0,\tilde r)}+H_+''(0,\tilde r).$$

We are almost done. All that is left is to collect terms and notice again $\int f''(s)ds=0$:

$$\begin{aligned}\lim_{\mu\to 0}\frac{\partial^2}{\partial\mu^2}\rho(\mu;r,p,\lambda) =& -\sum_{\tilde r}\left(4\lambda(\lambda-1)\frac{H_+'(0,\tilde r)^2}{H_+(0,\tilde r)}+H_+''(0,\tilde r)\right)\\ &-4\lambda(\lambda-1)\frac{H_+'(0,n)^2}{H_+(0,n)}-H_+''(0,n)\\ =& 4\lambda(1-\lambda)\left(\sum_{\tilde r}\frac{H_+'(0,\tilde r)^2}{H_+(0,\tilde r)}+\frac{H_+'(0,n)^2}{H_+(0,n)}\right).\end{aligned}$$

Putting this result together with the computations from the numerator above show the claim. □

B Posterior Visualization

In this section I show how the forces that impact the performance of different review systems can be viewed in posterior space. In particular, this visualization provides additional insight to the results of Section 3.2.

Let $f(s)=\frac{1}{2}\left(f_L(s)+f_H(s)\right)$ be the ex-ante distribution of signals under a symmetric prior. Moreover, let $\pi(s):=\frac{f_L(s)}{f_L(s)+f_H(s)}$ be the posterior belief placed on state L after observing signal s when there is a symmetric prior.

A straightforward computation shows that the learning efficiency under the full review is an expectation that depends on the posterior and the ex ante distribution:

$$\nu = 1 - \min_{\lambda \in [0,1]} 2 \int \pi(s)^{\lambda}(1-\pi(s))^{1-\lambda} f(s) ds.$$

Associated with each signal s is a unique posterior π. Let γ_π be the measure on $[0,1]$ that reflects the distribution of posteriors that is associated with f.[24] That is,

$$\gamma_\pi(B) := \frac{1}{2} \int_{\{s:\pi(s)\in B\}} f_L(s) + f_H(s) ds.$$

Then, the above can be rewritten as

$$\nu = 1 - \min_{\lambda \in [0,1]} 2 \int \pi^{\lambda}(1-\pi)^{1-\lambda} \gamma_\pi(d\pi). \tag{19}$$

This structure is preserved when review systems are introduced, as long as the review system treats posteriors consistently. That is, as long as it treats two signals that induce the same posterior the same. To this end, call a review system r *consistent* if, for all signals $s, s' \in \mathbb{R}$ such that $\pi(s) = \pi(s')$, then $r(s) = r(s')$.

The structure observed in (19) extends to consistent review systems. The difference is that instead of the function $\pi^\lambda(1-\pi)^{1-\lambda}$ being evaluated at each π, the review system r groups all posteriors that are mapped to the same review.

Proposition 4. *Let r be any consistent review system. Then,*

$$\nu(r) = 1 - \min_{\lambda \in [0,1]} 2 \int \left(\mathbb{E}_{\gamma_\pi}\left[\tilde{\pi} | r(\tilde{\pi}) = r(\pi)\right]\right)^{\lambda} \left(1 - \mathbb{E}_{\gamma_\pi}\left[\tilde{\pi} | r(\tilde{\pi}) = r(\pi)\right]\right)^{1-\lambda} \gamma_\pi(d\pi). \tag{20}$$

Proof. I show this in the case that r is a finite review system, and when γ^r has full support on R_r, since the notation is easier in this case. First, observe that $f_H(s) = f_L(s)\frac{1-\pi(s)}{\pi(s)}$. This means that we can write $\nu(r)$ as

$$1 - \nu(r) = \min_{\lambda \in [0,1]} \sum_{\tilde{r} \in R_r} \mathbb{P}_L(r^{-1}(\tilde{r}))^{\lambda} \mathbb{P}_H(r^{-1}(\tilde{r}))^{1-\lambda}$$

[24]Although γ_π is the measure of posteriors on state L, I use γ_π to indicate that this is the ex-ante measure, and not the measure conditional on state L.

$$= \min_{\lambda\in[0,1]} \sum_{\tilde{r}\in R_r} \left(\int_{\{s\in r^{-1}(\tilde{r})\}} \frac{\pi(s)}{(1-\pi(s))} f_H(s)ds \right)^{\lambda} \cdot \left(\int_{\{s\in r^{-1}(\tilde{r})\}} f_H(s)ds \right)^{1-\lambda}$$

Now, if $\pi(s)$ is injective, it is the case (by straightforward computation) that $\frac{1}{1-\pi} f_H(s) = 2\gamma_\pi(d\pi(s))$ and so also $f_H(s) = 2(1-\pi(s))\gamma_\pi(d\pi(s))$. In general (even if $\pi(s)$ is not injective), for each λ, then,

$$\begin{aligned}
&\sum_{\tilde{r}\in R_r} \mathbb{P}_L(r^{-1}(\tilde{r}))^{\lambda} \mathbb{P}_H(r^{-1}(\tilde{r}))^{1-\lambda} \\
&\quad = \sum_{\tilde{r}\in R_r} 2 \left(\int_{\tilde{\pi}\in r^{-1}(\tilde{r})} \tilde{\pi}\gamma_\pi(d\tilde{\pi}) \right)^{\lambda} \cdot \left(\int_{\tilde{\pi}\in r^{-1}(\tilde{r})} (1-\tilde{\pi})\gamma_\pi(d\tilde{\pi}) \right)^{1-\lambda} \\
&\quad = \sum_{\tilde{r}\in R_r} 2 \frac{\left(\int_{\tilde{\pi}\in r^{-1}(\tilde{r})} \tilde{\pi}\gamma_\pi(d\tilde{\pi}) \right)^{\lambda} \cdot \left(\int_{\tilde{\pi}\in r^{-1}(\tilde{r})} (1-\tilde{\pi})\gamma_\pi(d\tilde{\pi}) \right)^{1-\lambda}}{\int_{\tilde{\pi}\in r^{-1}(\tilde{r})} \gamma_\pi(d\tilde{\pi})} \int_{\tilde{\pi}\in r^{-1}(\tilde{r})} \gamma_\pi(d\tilde{\pi}) \\
&\quad = \sum_{\tilde{r}\in R_r} 2 \left(\mathbb{E}_{\gamma_\pi}[\pi | r(\pi) = \tilde{r}] \right)^{\lambda} \left(1 - \mathbb{E}_{\gamma_\pi}[\pi | r(\pi) = \tilde{r}] \right)^{1-\lambda} \int_{\tilde{\pi}\in r^{-1}(\tilde{r})} \gamma_\pi(d\tilde{\pi})
\end{aligned}$$

From here, we minimize over λ, and conclude. □

Notice that here the measure with respect to which the integral is taken is simply $\gamma_\pi(d\pi)$, which is *independent* of r. This means that, fixing λ, the loss in information can be thought of as coming exactly from the pre-evaluation of the function $\pi^{\lambda}(1-\pi)^{1-\lambda}$. Since, for any λ, this function is concave, information loss is exactly due to Jensen's inequality. The interpretation is exactly the same as the intuition as before: it is more efficient to separate an informative signal and an uninformative signal than it is to have them grouped into a moderately informative signal. Proposition 4 gives a way to exactly quantify this trade-off.

This analysis can be used to explain why increases in homogeneity increase the asymmetry of the optimal review system. In particular, the larger the spread of the posteriors within a bin, the more information will be lost within that bin, because the larger will be the difference in Jensen's inequality. For small values of μ, different values of α lead to very different distributions of posteriors $d\gamma_\pi$. As observed in Figure 1a, as α increases, simultaneously more signals are very uninformative and more signals are very informative. This is confirmed in Figure 1b, where the distributions of posteriors are plotted for the same values of α. As α increases, the spread in the distribution of posteriors increases, driving the increase in $\kappa_\alpha(0)$. At the same time, the larger mass near certainty (mass near 0 and 1) incentivizes moving the threshold away from symmetry.

B.1 Optimality of Threshold Rules

If r is a consistent review system that has b bins, then it is effectively a partition of the space $[0,1]$. Let $\{B_1, ..., B_b\}$ be the partition of the interval. Then (20) can be rewritten as

$$\nu(r) = 1 - \min_{\lambda \in [0,1]} 2 \sum_{i=1}^{b} \left(\mathbb{E}_{\gamma_\pi}\left[\pi | \pi \in B_i\right]\right)^\lambda \left(1 - \mathbb{E}_{\gamma_\pi}\left[\pi | \pi \in B_i\right]\right)^{1-\lambda} \gamma_\pi\left(B_i\right).$$

With this structure, it is possible to show the intuitive result that optimal coarsening with a fixed number of bins must necessarily be threshold rules in posterior space. First, I explicitly define what it means to be defined by thresholds.

Definition 5. A review system r is called a *b-threshold* review function if $r^{-1}(\tilde{r})$ is convex (except perhaps on a set of measure zero) for each $\tilde{r} \in R_r$.

Optimal review systems are necessarily threshold rules.

Proposition 5. *Let r be an optimal review system with b bins. Then r is b-threshold review function.*

Proof. I show that if r is a finite review with b bins that is not such a threshold rule, then there exists a review with b bins that outperforms it. Moreover, I show that this holds point-wise for each λ, so that certainly it holds over the minimum of λ.

Consider two bins r_1 and r_2 that cannot be separated by any threshold. Let $\pi_i = \mathbb{E}[\pi | r(\pi) = r_i]$. Without loss assume that $\pi_1 \leq \pi_2$.

Consider any point $\overline{\pi}$ that lies between π_1 and π_2 (if they are equal, then $\overline{\pi} = \pi_1 = \pi_2$). By assumption, $\overline{\pi}$ is not a threshold for these two bins. In particular, this means that because $\pi_1 \leq \overline{\pi} \leq \pi_2$, there exists a set Π_1 with positive measure such that for all $\pi \in \Pi_1$, $\pi > \overline{\pi}$ and $r(\pi) = r_1$. That is, consider the following two sets

$$\Pi_1 := \{\pi | r(\pi) = r_1 \text{ and } \pi > \overline{\pi}\} \quad \text{and}$$
$$\Pi_2 := \{\pi | r(\pi) = r_2 \text{ and } \pi < \overline{\pi}\}.$$

Because $\overline{\pi}$ is not a threshold for r_1, r_2, and $\pi_1 \leq \overline{\pi} \leq \pi_2$, these sets both must have positive measure. Suppose that both of their measures are larger than ϵ. Select a subset of each that has mass ϵ: $\Pi_1^\epsilon, \Pi_2^\epsilon$. Let $\pi_i' = \mathbb{E}[\pi | r'(\pi) = r_i]$ denote the new conditional expectations. Because of the construction it must necessarily be the case that $\pi_1 > \pi_1'$ and $\pi_2 < \pi_2'$.

Now, define a new review function r' as follows:

$$r'(\pi) = \begin{cases} r(\pi) & \text{if} \quad \pi \notin \Pi_1^\epsilon, \Pi_2^\epsilon \\ r_1 & \text{if} \quad \pi \in \Pi_2^\epsilon \\ r_2 & \text{if} \quad \pi \in \Pi_1^\epsilon \end{cases}.$$

That is, r' agrees with r, except that it swaps the Π_i^ϵ's. I show that we necessarily have $\nu(r') \geq \nu(r)$. As mentioned, I show that this holds for each λ. To that end, fix a value of $\lambda \in (0, 1)$, and let $\varphi(\pi) := \pi^\lambda (1 - \pi)^{1-\lambda}$. I now show that the iterated expectation of $\varphi(\pi)$ on the partition induced by r' is lower than on the partition induced by r.

That is, I aim to show that

$$\sum_{i=1}^{b} \varphi\left(\mathbb{E}_{\gamma_\pi}\left[\pi | r(\pi) \in r_i\right]\right) \gamma_\pi(r^{-1}(r_i)) \geq \sum_{i=1}^{b} \varphi\left(\mathbb{E}_{\gamma_\pi}\left[\pi | r'(\pi) \in r_i\right]\right) \gamma_\pi((r')^{-1}(r_i)).$$

Notice now that the two agree except on r_1 and r_2, so that the sum simplifies. Now, consider adding a linear function $h(\pi)$ to $\varphi(\pi)$ such that the unique maximum of $h + \varphi$ is obtained at $\overline{\pi}$. Note that this is possible because φ is strictly concave. In particular, this means that $h+\varphi$ is strictly increasing for $\pi < \overline{\pi}$ and strictly decreasing for $\pi > \overline{\pi}$. Then, because h is linear, it is the case that

$$\begin{aligned} & \sum_{i=1}^{2} \varphi\left(\pi_i\right) \gamma_\pi(r^{-1}(r_i)) \geq \sum_{i=1}^{2} \varphi\left(\pi_i'\right) \gamma_\pi((r')^{-1}(r_i)) \\ \iff \quad & \sum_{i=1}^{2} \left(\varphi\left(\pi_i\right) + h(\pi_i)\right) \gamma_\pi(r^{-1}(r_i)) \geq \sum_{i=1}^{2} \left(\varphi\left(\pi_i'\right) + h(\pi_i')\right) \gamma_\pi((r')^{-1}(r_i)). \end{aligned}$$

As observed, it is the case that $\overline{\pi} \geq \pi_1 > \pi_1'$. This means that

$$\varphi\left(\pi_1\right) + h(\pi_1) > \varphi\left(\pi_1'\right) + h(\pi_1')$$

It is similarly the case that

$$\varphi\left(\pi_2\right)+h(\pi_2)>\varphi\left(\pi_2'\right)+h(\pi_2')$$

Since by construction it is the case that $\gamma_\pi(r^{-1}(r_i))=\gamma_\pi((r')^{-1}(r_i))$ for $i=1,2$, this means that we have achieved a strict improvement. Hence it must be the case that $\nu(r)<\nu(r')$. □

C Relationship between the Platform and Consumers' Preference over Information Structures

In this section, I study the perspective of a consumer choosing which source of information to use when making their decision. In practice, many platforms elicit multiple types of information from reviewers; often one can leave a coarse rating and then supplement the rating with a free-text full review.

In the work above I take the perspective of the platform ex ante choosing what type of review to elicit. In that choice the platform internalizes the randomness in the review decisions of reviewers. If there are multiple types of reviews that have been collected, when it comes time for consumers to choose between them, this randomness has been removed. So, the comparison is not between the rates of acquisition, but directly between the number of reviews of each type of review.

The comparison from the side of consumers is more similar to asking how many coarsened reviews "equals" one review of another type. The distinction between the two is subtle, but as shown below, the removal of risk leads the consumer to require more coarsened reviews. However, as individual reviews become uninformative, this difference disappears and so the metric of learning loss from Theorem 2 applies again.

Lemma 8. *Fix a consumer with a (finite) decision problem, two review systems r, r', and a degree of distinction μ. Then there exists an N large such that for all $n_r, n_{r'}>N$, n_r reviews of type r is preferred to $n_{r'}$ reviews of type r' if*

$$\frac{n_r}{n_{r'}}>\frac{\log(1-\nu(\mu;r'))}{\log(1-\nu(\mu;r))}.$$

Proof. Let (n_r, r) denote the statistical experiment that generates n_r conditionally independent signals of drawn according to r. Then, because each of the signals are independent,

$$
\begin{aligned}
\nu(\mu;(n_r,r)) =& 1-\min_{\lambda\in[0,1]}\int_R \dots \int_R \prod_{i=1}^{n_r}\left(\gamma_L^r(d\tilde{r}_i)\right)^{\lambda}\left(\gamma_H^r(d\tilde{r}_i)\right)^{1-\lambda} \\
=& 1-\min_{\lambda\in[0,1]}\prod_{i=1}^{n_r}\int_R \left(\gamma_L^r(d\tilde{r}_i)\right)^{\lambda}\left(\gamma_H^r(d\tilde{r}_i)\right)^{1-\lambda} \\
=& 1-(1-\nu(\mu;r))^{n_r}.
\end{aligned}
$$

From here, the result comes from an application of either Torgersen's theorem or Theorem 1. Note that this result follows the same logic as Proposition 3 in Mu et al. (2021). See also Chernoff (1952), which discusses this relationship at some length. $\square$

This condition is similar to, but not exactly, the condition of Theorem 1. As the next Proposition highlights, because the information for consumers is not random, consumers place more weight on "good" information than the platform. However, as information becomes very noisy, this difference shrinks and eventually disappears in the limit. Specifically, the relationship is as follows.

Proposition 6. *Let* $\kappa(\mu;r',r) := \frac{\nu(\mu;r')}{\nu(\mu;r)}$ *and let* $\zeta(\mu;r',r) := \frac{\log(1-\nu(\mu;r'))}{\log(1-\nu(\mu;r))}$. *Then,*

$$
\kappa(\mu;r',r) < \zeta(\mu;r',r) \iff \nu(\mu;r') > \nu(\mu;r).
$$

However,

$$
\lim_{\mu\to 0}\kappa(\mu;r',r) = \lim_{\mu\to 0}\zeta(\mu;r',r).
$$

Proof. For all $x, y \in (0,1)$,

$$
\frac{\log(1-x)}{\log(1-y)} > \frac{x}{y} \iff x > y.
$$

This shows the first claim. For the second, one can apply l'Hopital's rule to the limit of $\mu \to 0$ for

$$
\zeta(\mu;r',r) = \frac{\log(1-\nu(\mu;r'))}{\log(1-\nu(\mu;r))}.
$$

and find that

$$\lim_{\mu \to 0} \zeta(\mu; r', r) = \lim_{\mu \to 0} \frac{\frac{\partial}{\partial \mu} \nu(\mu; r')}{\frac{\partial}{\partial \mu} \nu(\mu; r)} = \lim_{\mu \to 0} \eta(\mu; r', r).$$

This shows the second claim, completing the proof. □

The platform's choice between ratings systems must necessarily reflect the randomness with which it receives information. This means that, relative to a consumer who does not need to factor-in this randomness, the platform will value more frequent, but uninformative, reviews. This result suggests why in practice platforms collect multiple sources of information: consumers trade-off between sources of information in different ways than do platforms, because of the timing of their choice between sources of information.